Exam Preparation

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Analysis from the Practice Sample questions

- Exam Page: https://aws.amazon.com/certification/certified-solutions-architect-professional/

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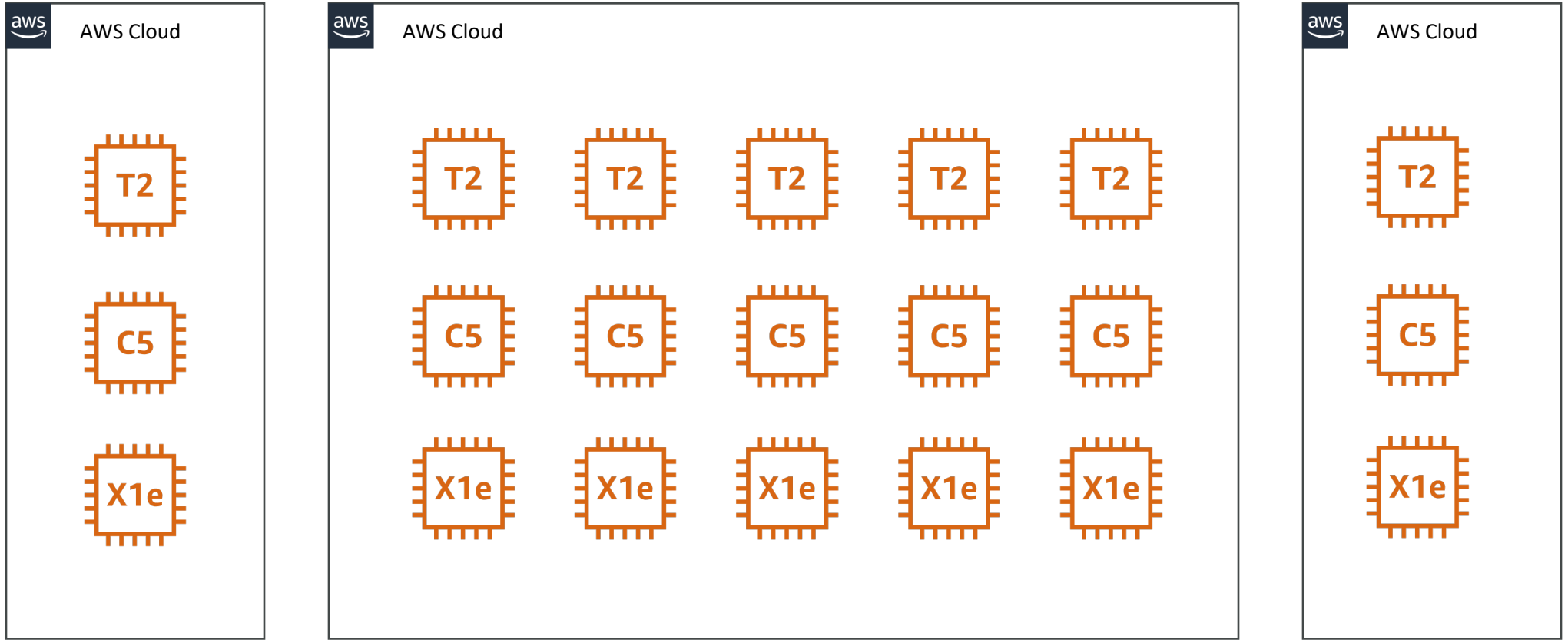
Question 1

- An enterprise has a large number of AWS accounts owned by separate business groups. One of the accounts was recently compromised. The attacker launched a large number of instances, resulting in a high bill for that account.
- The security breach was addressed, but management has asked a solutions architect to develop a solution to **help prevent excessive spending** in all accounts. **Each business group wants to retain full control over its AWS account.**
- Which solution should the solutions architect recommend to meet these requirements?

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Question 1 – Architecture Diagram

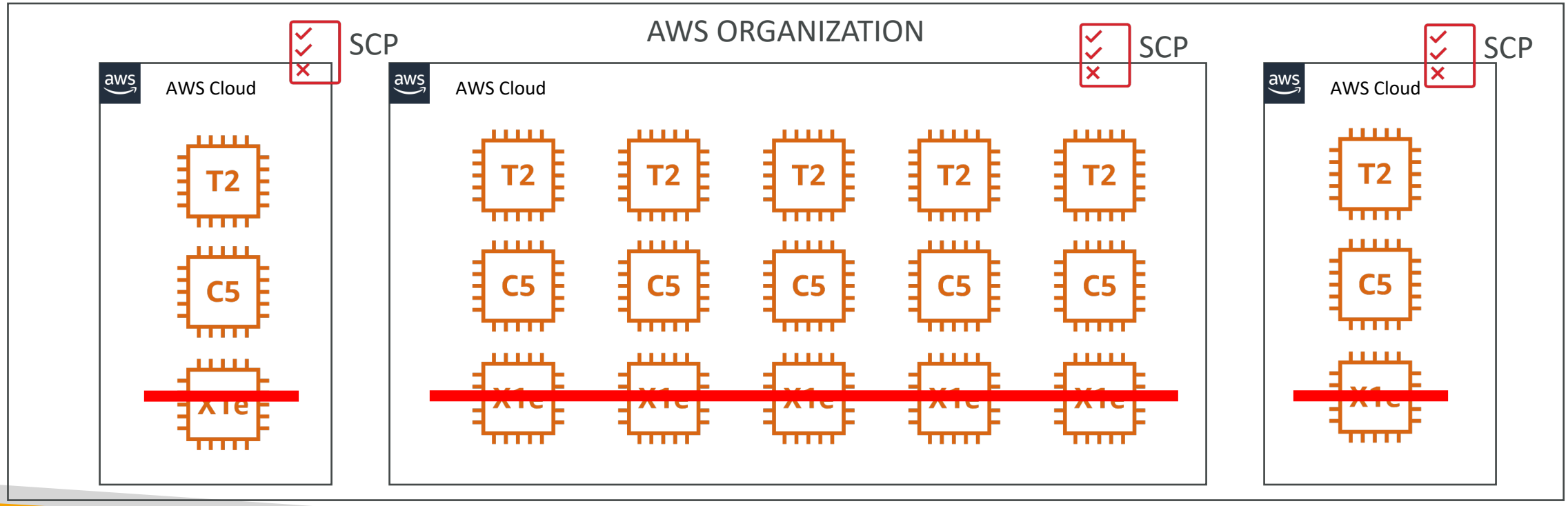


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Option A

- Use AWS Organizations to add each AWS account to the master account. Create a service control policy (SCP) that uses the `ec2:instanceType` condition key to prevent the launch of high-cost instance types in each account.

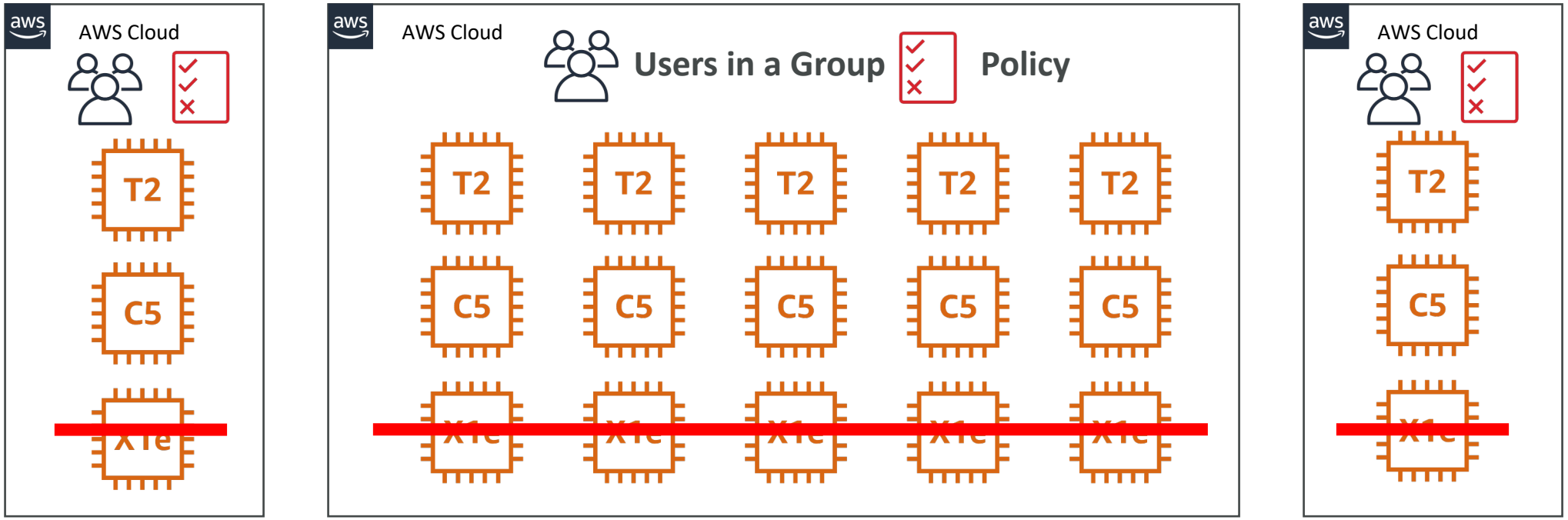


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Option B

- Attach a new customer-managed IAM policy to an IAM group in each account that uses the ec2:instanceType condition key to prevent the launch of high-cost instance types. Place all of the existing IAM users in each group.



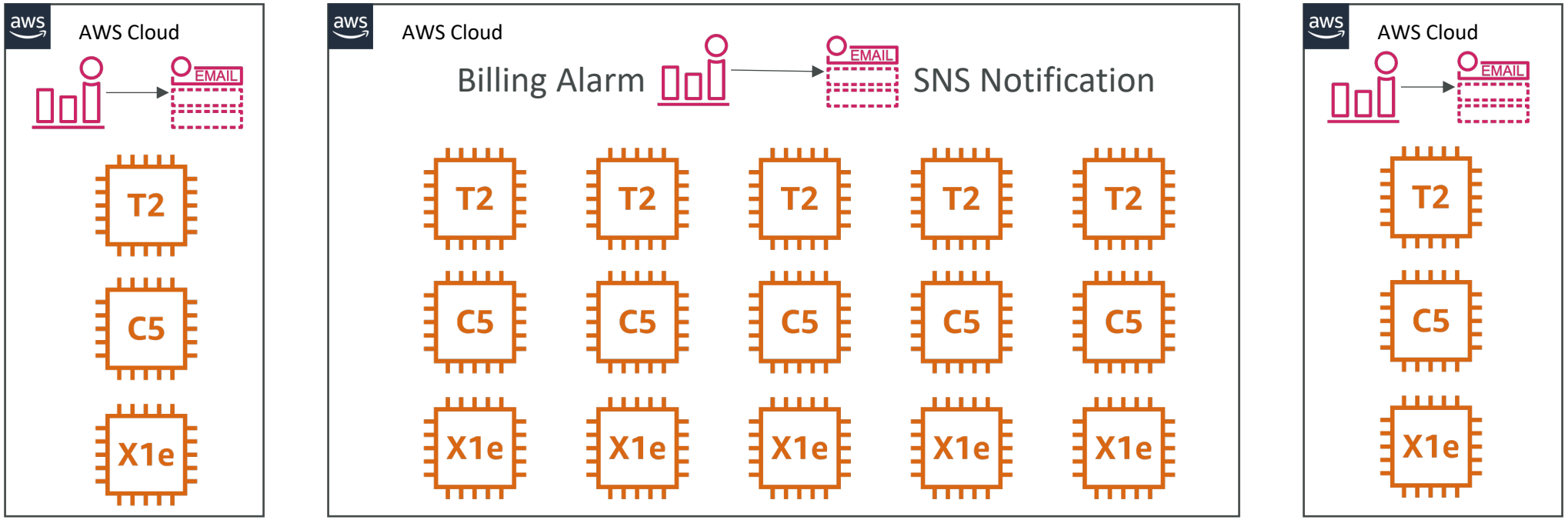
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Option C

CORRECT ANSWER

- Enable billing alerts on each AWS account. Create Amazon CloudWatch alarms that send an Amazon SNS notification to the account administrator whenever their account exceeds the spending budget.

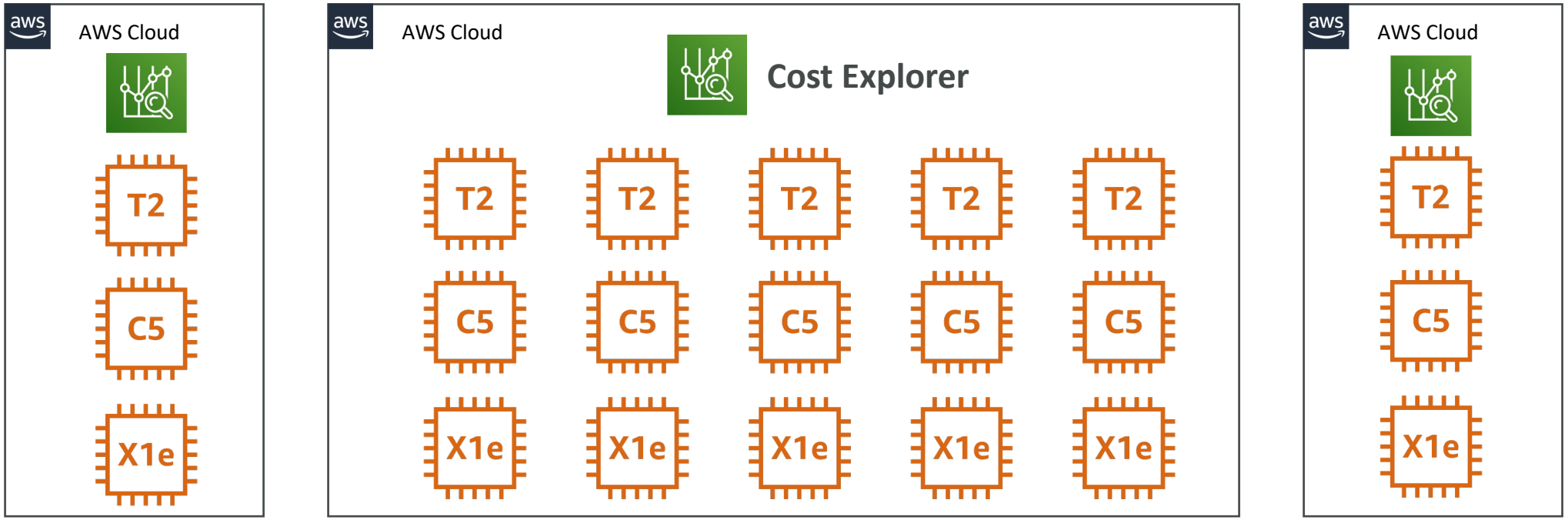


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Option D

- Enable AWS Cost Explorer in each account. Regularly review the Cost Explorer reports for each account to ensure spending does not exceed the planned budget



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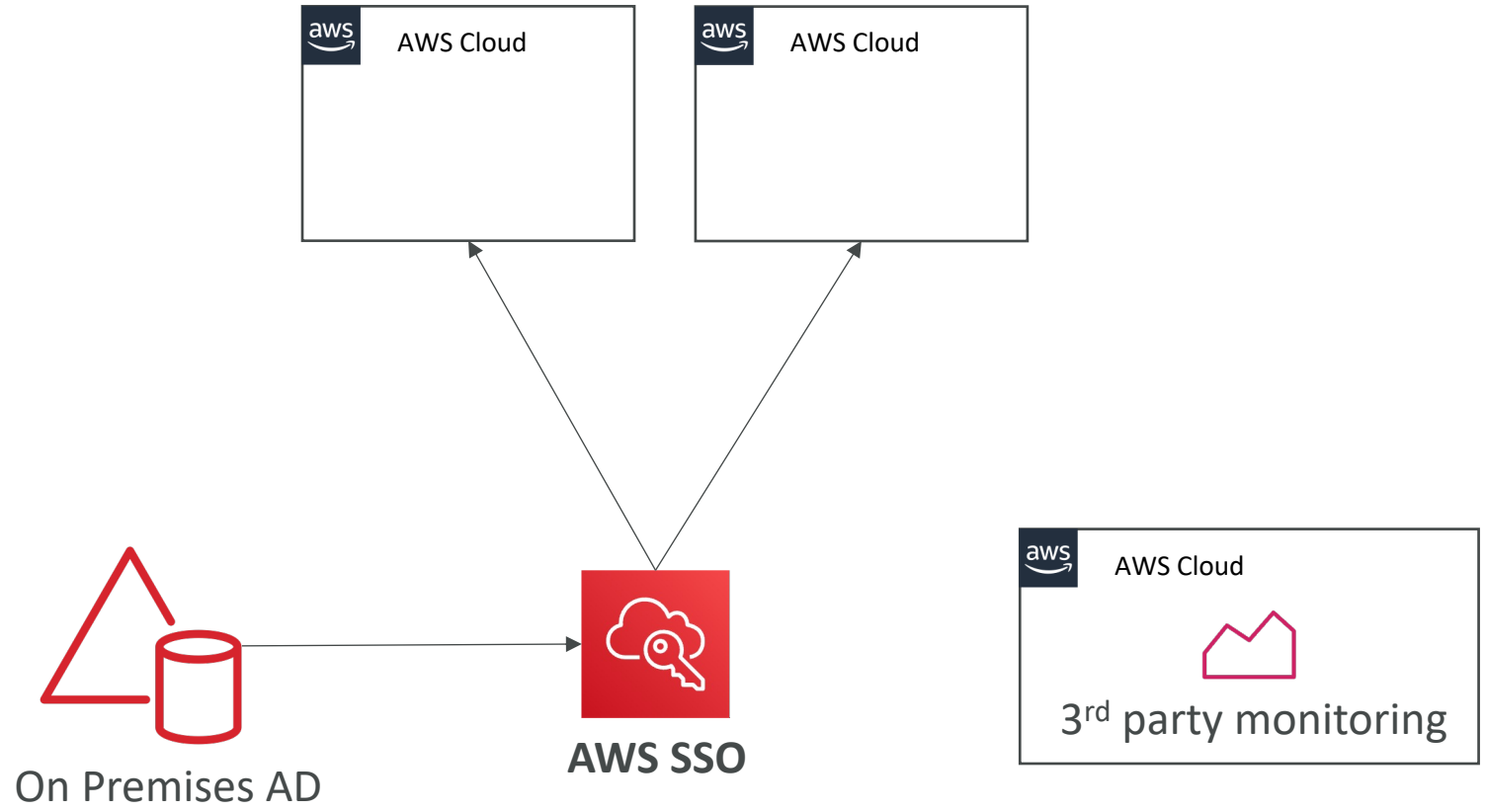
Question 2

- A company has multiple AWS accounts. The company has integrated its on-premises Active Directory (AD) with **AWS SSO to grant AD users least privilege abilities to manage infrastructure across all the accounts.**
- A solutions architect must integrate a third-party monitoring solution that requires **read-only access across all AWS accounts.** The monitoring solutions will run **in its own AWS account.**
- How can the monitoring solution be given the required permissions?

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Question 2 - Architecture

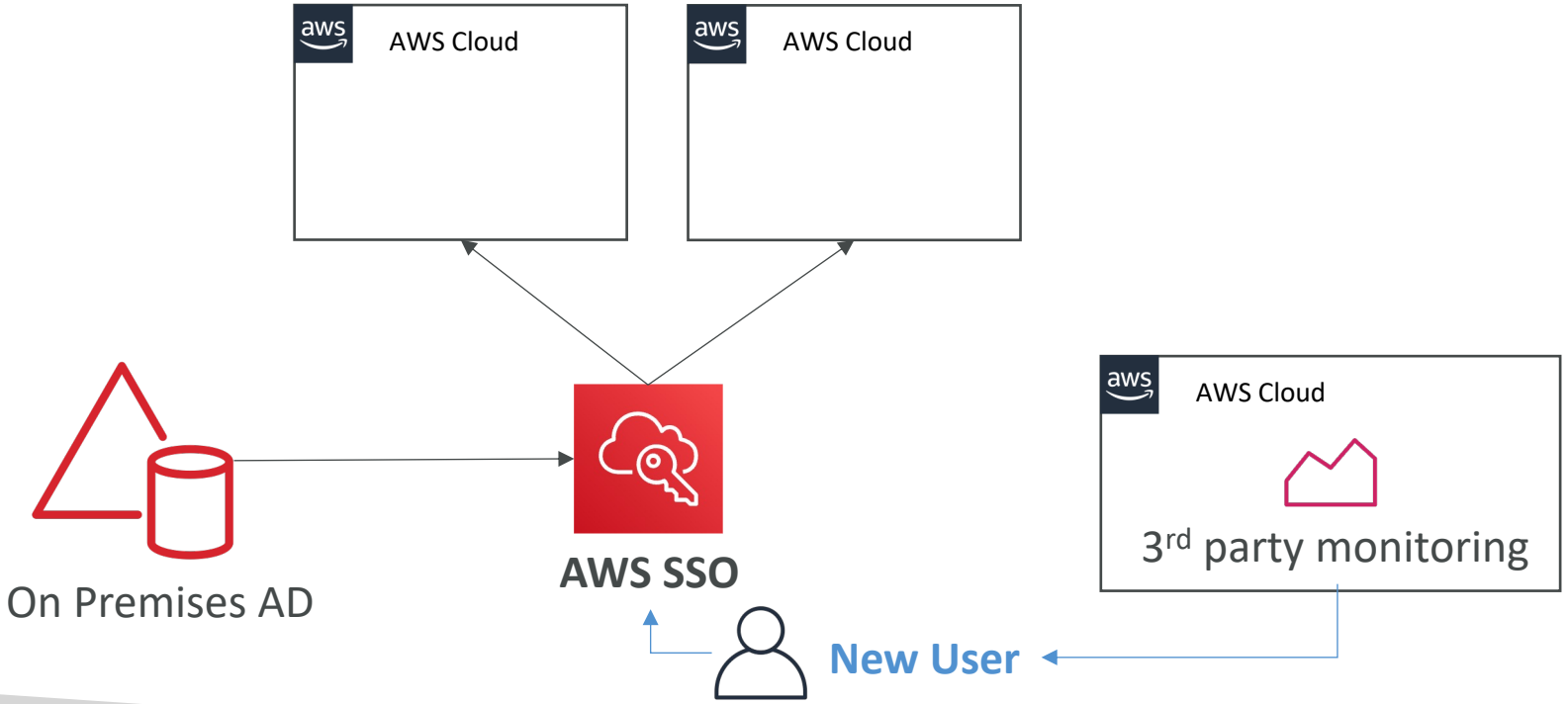


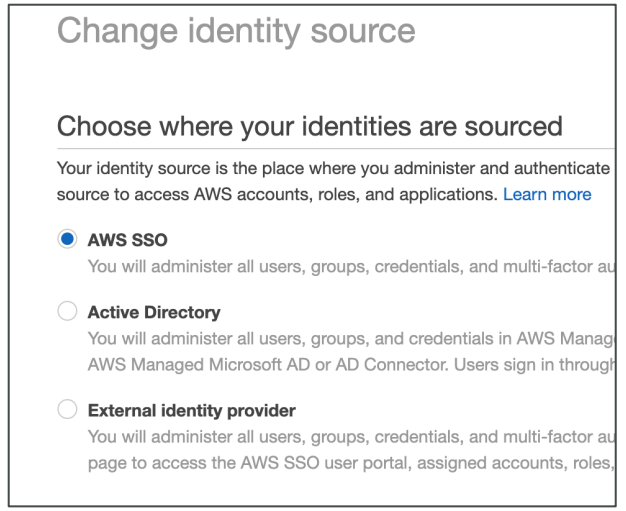
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Option A

- Create a user in an AWS SSO directory and assign a read-only permissions set. Assign all AWS accounts to be monitored to the new user. Provide the third-party monitoring solution with the user name and password.





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Note on option A

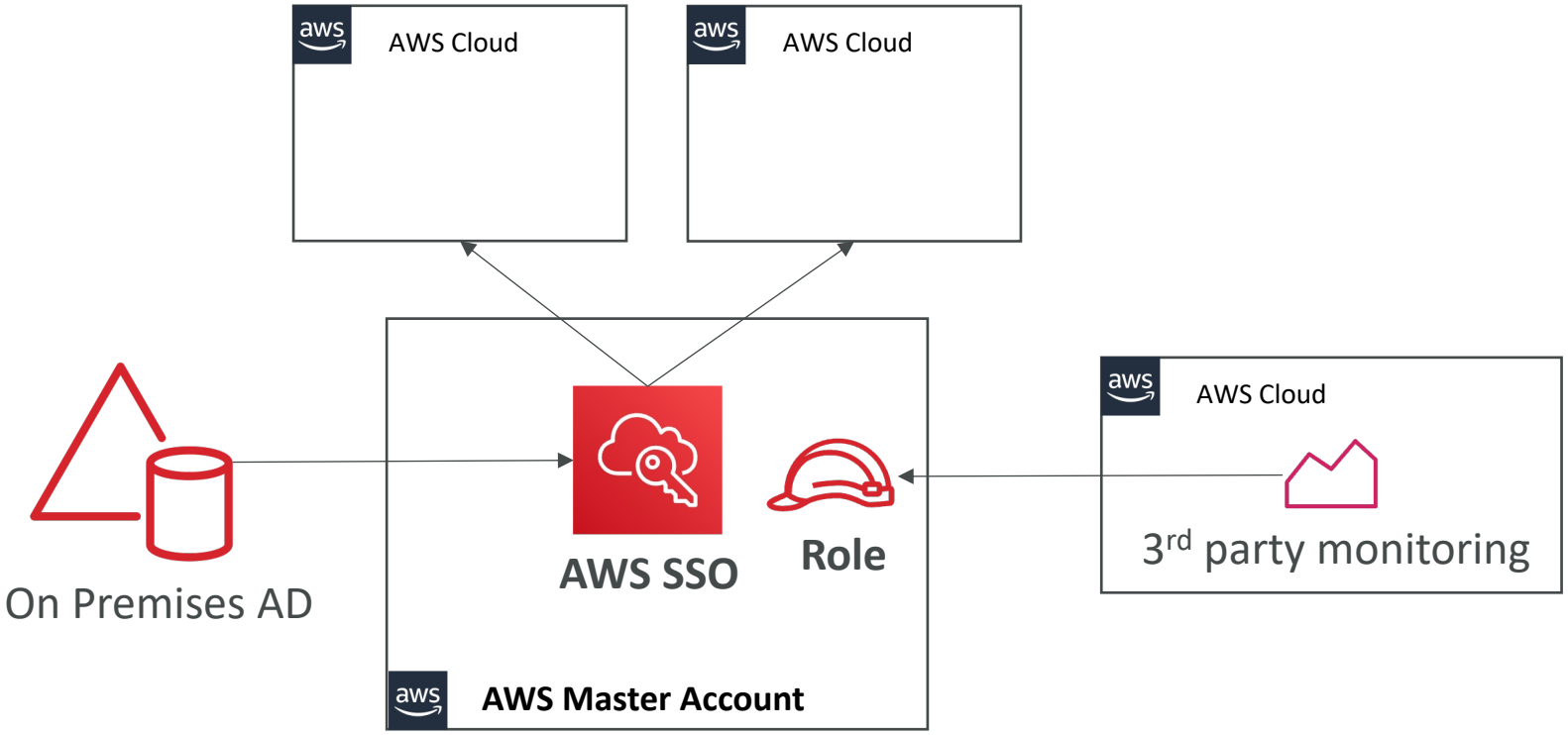
- Currently, the sample question PDF says:
- "A is incorrect **because credentials supplied by AWS SSO are temporary**, so the application would lose permissions and have to re-login"
- That is **wrong**.
- Users created in AWS SSO have a password that doesn't change and must respect the password policy defined.
- Here Option A is wrong because you can't have both users defined in AWS SSO and in Active Directory. SSO only allows for one Identity source (SSO, AD or IdP).

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Option B

- Create an AWS IAM role in the organization's master account. Allow the AWS account of the third-party monitoring solution to assume the role.

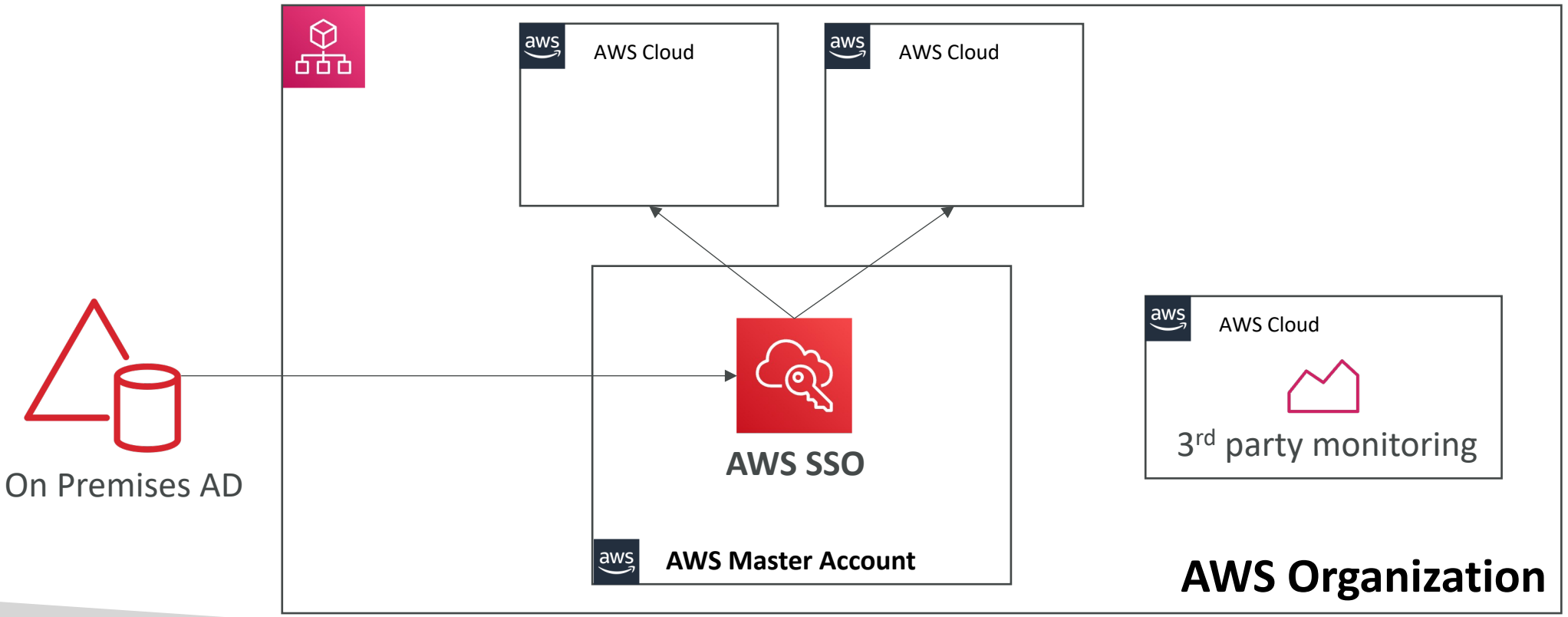


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Option C

- Invite the AWS account of the third-party monitoring solution to join the organization. Enable all features



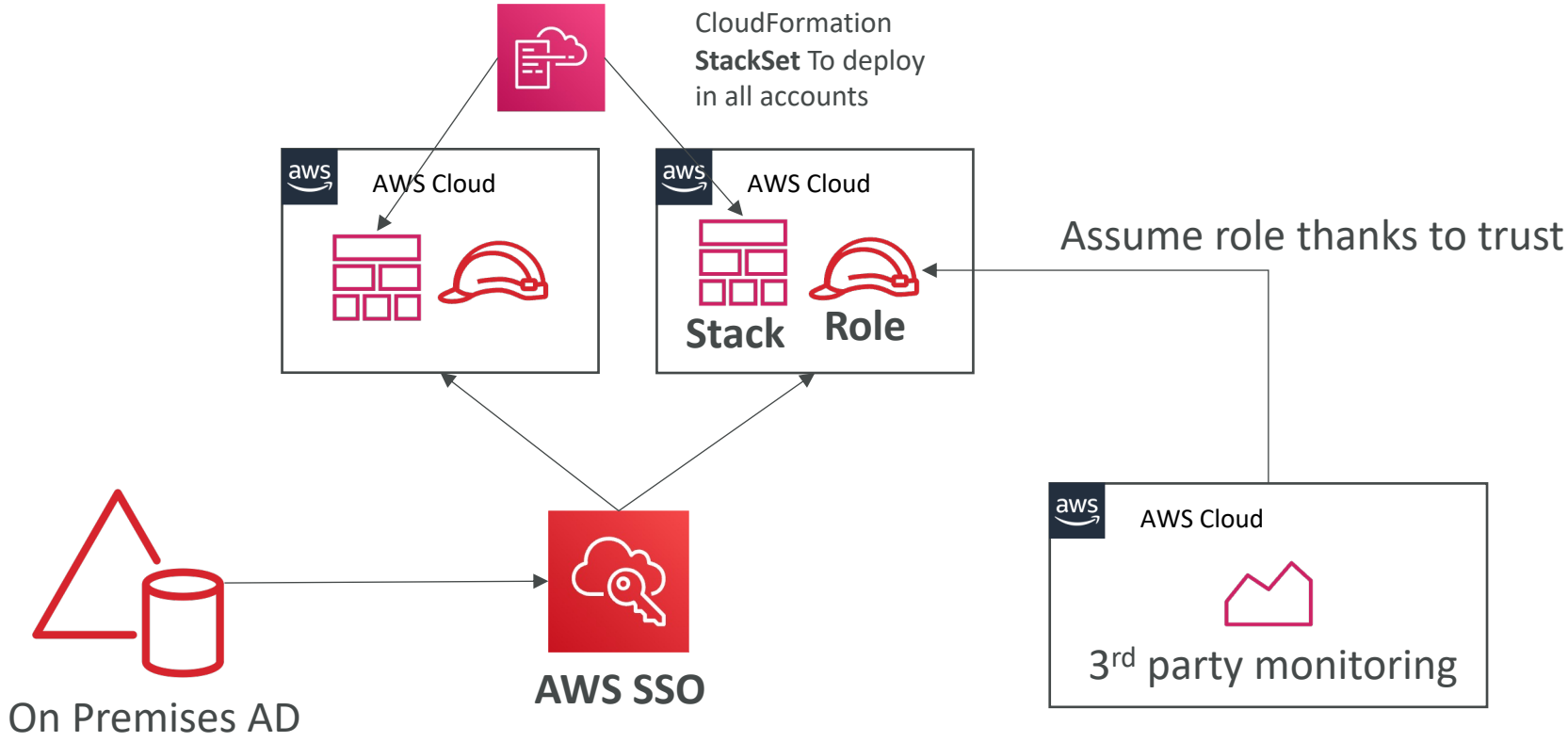
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Option D

CORRECT ANSWER

- Create an AWS CloudFormation template that defines a new AWS IAM role for the third-party monitoring solution with the account of the third party listed in the trust policy. Create the IAM role across all linked AWS accounts by using a stack set.



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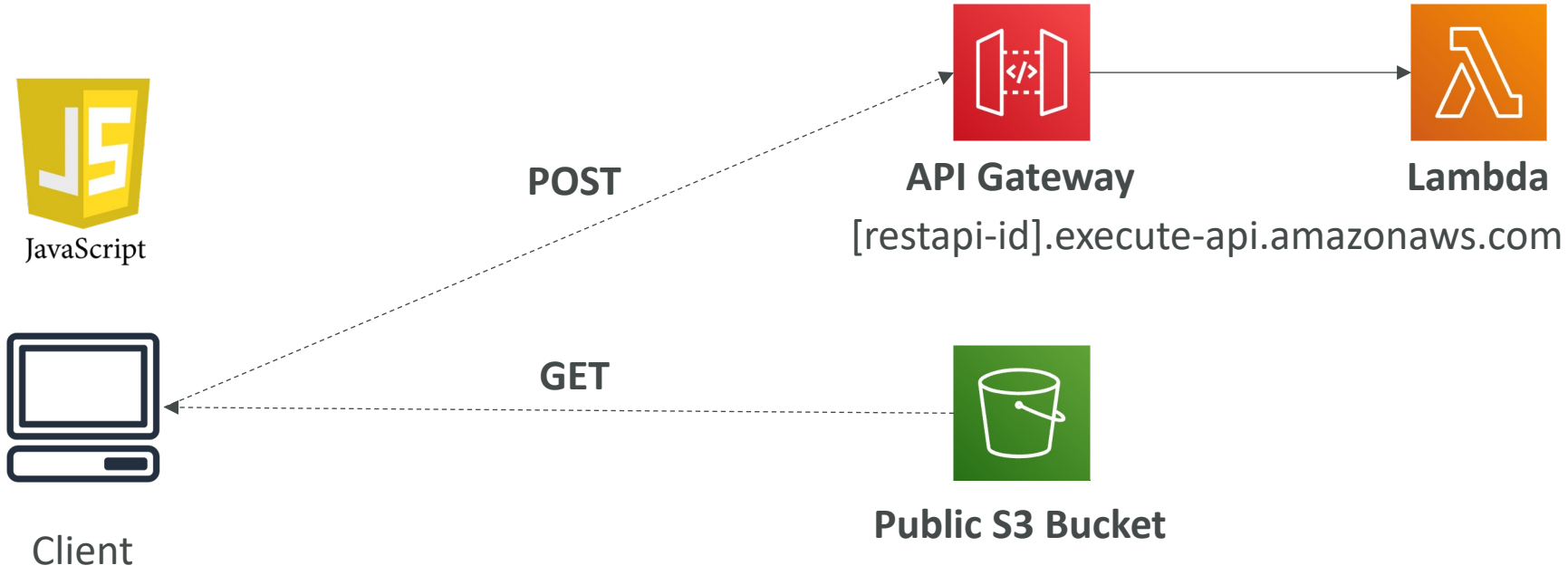
Question 3

- A team is building an **HTML form** hosted in a **public Amazon S3 bucket.** The form uses **JavaScript to post data to an Amazon API Gateway** endpoint. The endpoint is integrated with AWS Lambda functions. The team has tested each method in the API Gateway console and received valid responses.
- Which combination of steps must be completed for the form **to successfully post to the API Gateway** and receive a valid response? (Select TWO.)

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Question 3 – Architecture



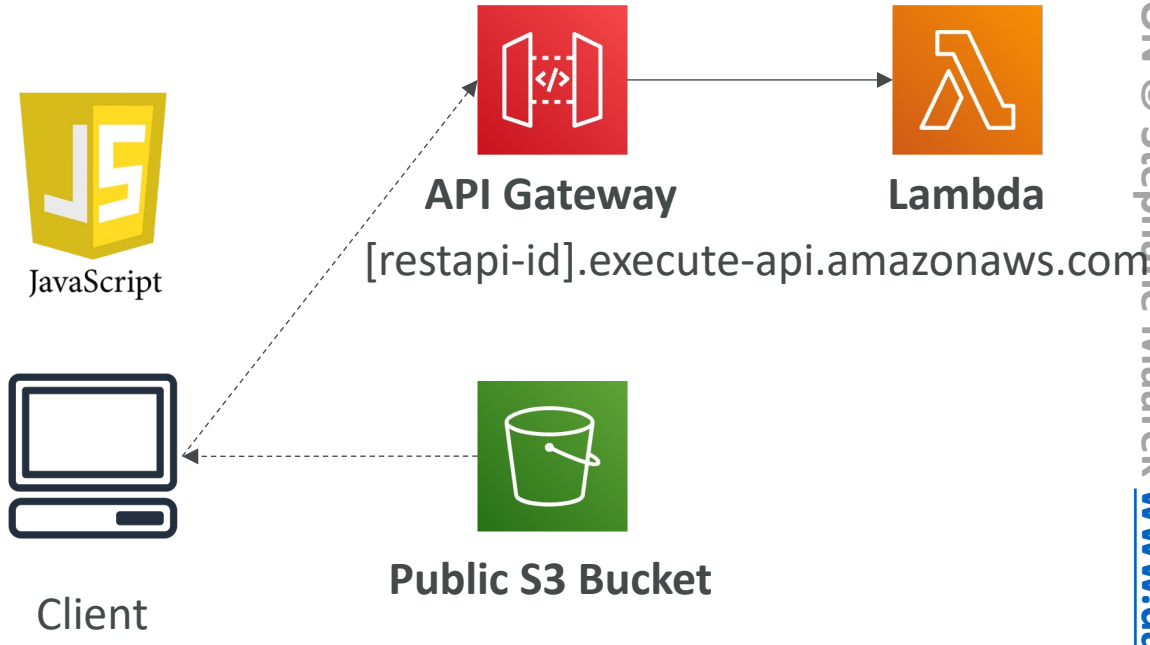
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Options

CORRECT ANSWERS D,E

- A) Configure the S3 bucket to allow cross-origin resource sharing (CORS).
- B) Host the form on Amazon EC2 rather than Amazon S3.
- C) Request a limit increase for API Gateway.
- D) Enable cross-origin resource sharing (CORS) in API Gateway.
- E) Configure the S3 bucket for web hosting.

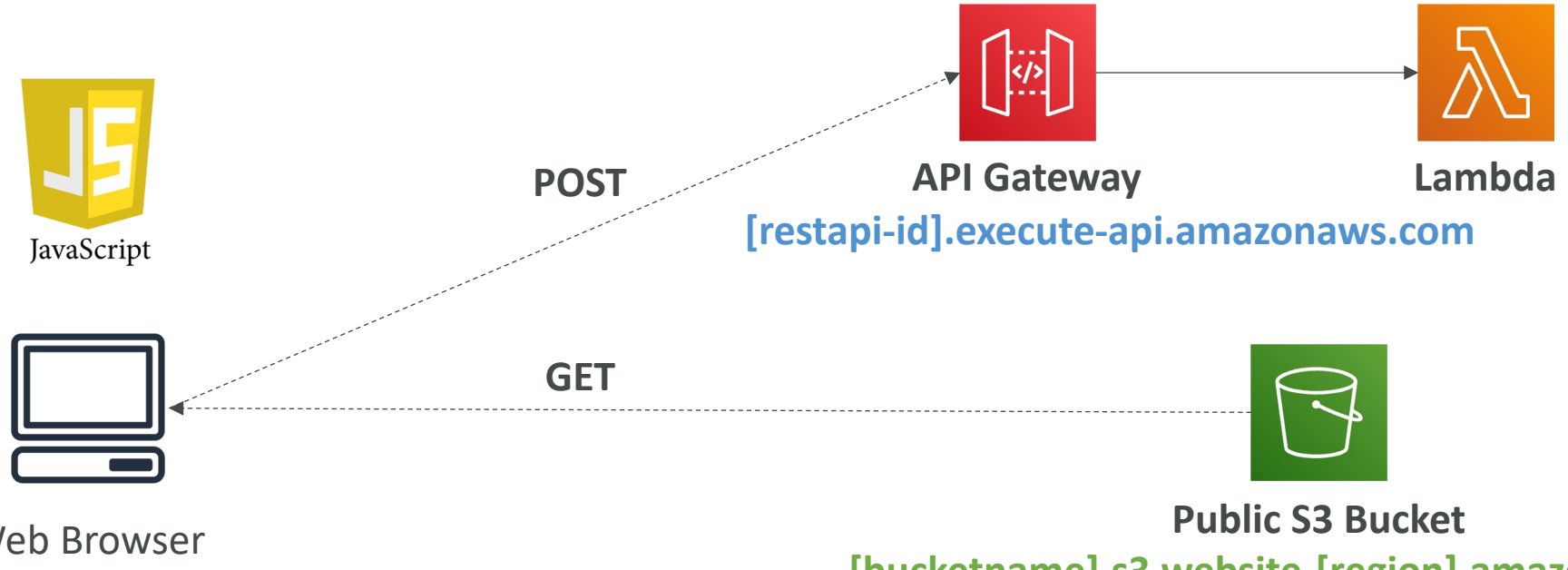


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Question 3 – Final Architecture
CORS is a Browser based security

CORS to allow calls with Origin **[bucketname].s3.website-[region].amazonaws.com**
Using the header **Access-Control-Allow-Origin**



Visits [bucketname].s3.website-[region].amazonaws.com
Makes API calls to [restapi-id].execute-api.amazonaws.com
With **Origin:** [bucketname].s3.website-[region].amazonaws.com

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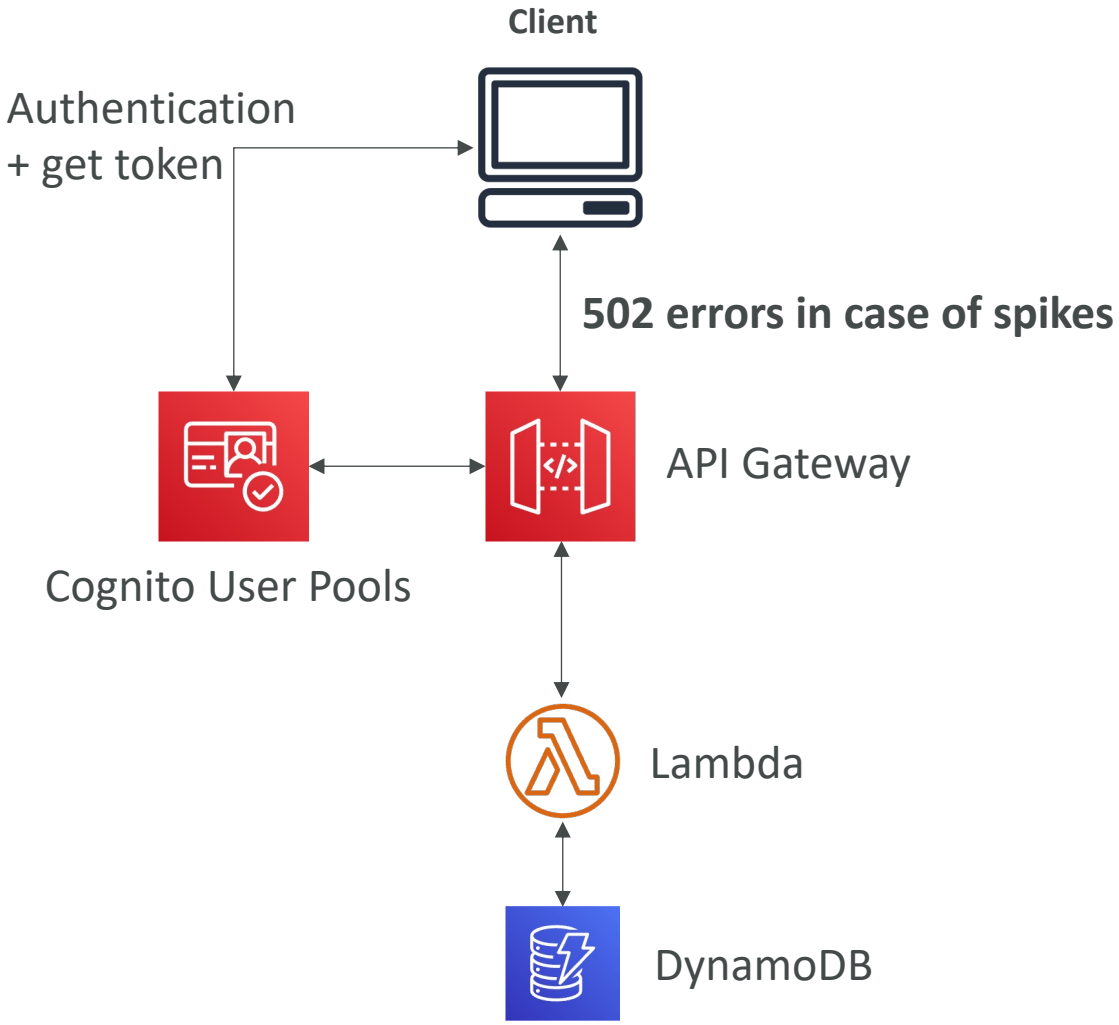
Question 4

- A retail company runs a serverless mobile app built on Amazon API Gateway, AWS Lambda, Amazon Cognito, and Amazon DynamoDB. During **heavy holiday traffic spikes**, the company receives complaints of **intermittent system failures**. Developers find that the API Gateway endpoint is returning **502 Bad Gateway errors to seemingly valid requests.**
- Which method should address this issue?

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Question 4 – Architecture



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API Gateway - Errors

- 4xx means Client errors
  - 400: Bad Request
  - 403: Access Denied, WAF filtered
  - 429: Quota exceeded, Throttle

- 5xx means Server errors
  - **502: Bad Gateway Exception, usually for an incompatible output returned from a Lambda proxy integration backend and occasionally for out-of-order invocations due to heavy loads.**
  - 503: Service Unavailable Exception
  - 504: Integration Failure – ex Endpoint Request Timed-out Exception
    API Gateway requests time out after 29 second maximum

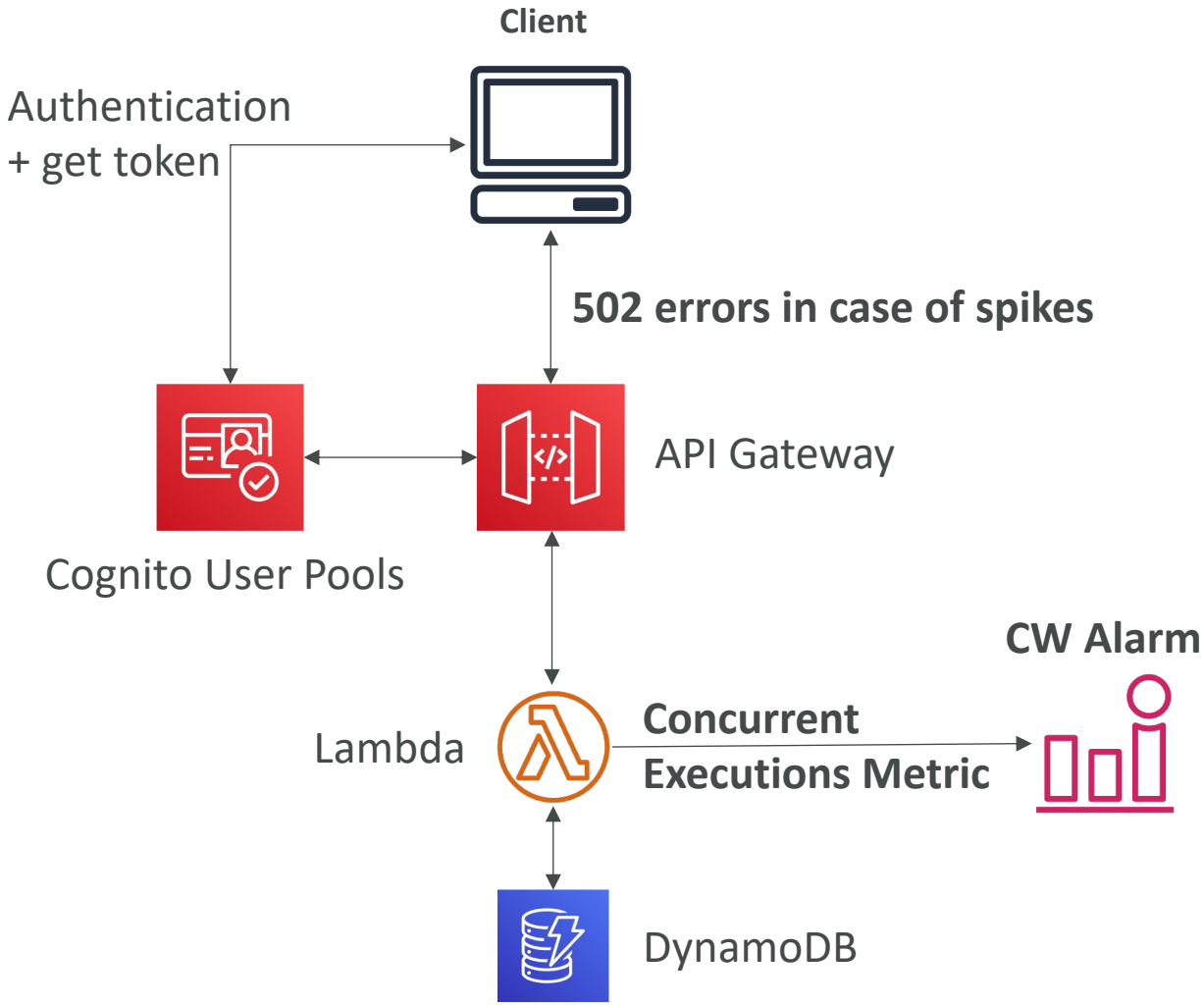
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Option A

CORRECT ANSWER

- Increase the **concurrency limit for Lambda functions** and configure notification alerts to be sent by Amazon CloudWatch when the **ConcurrentExecutions** metric approaches the limit.



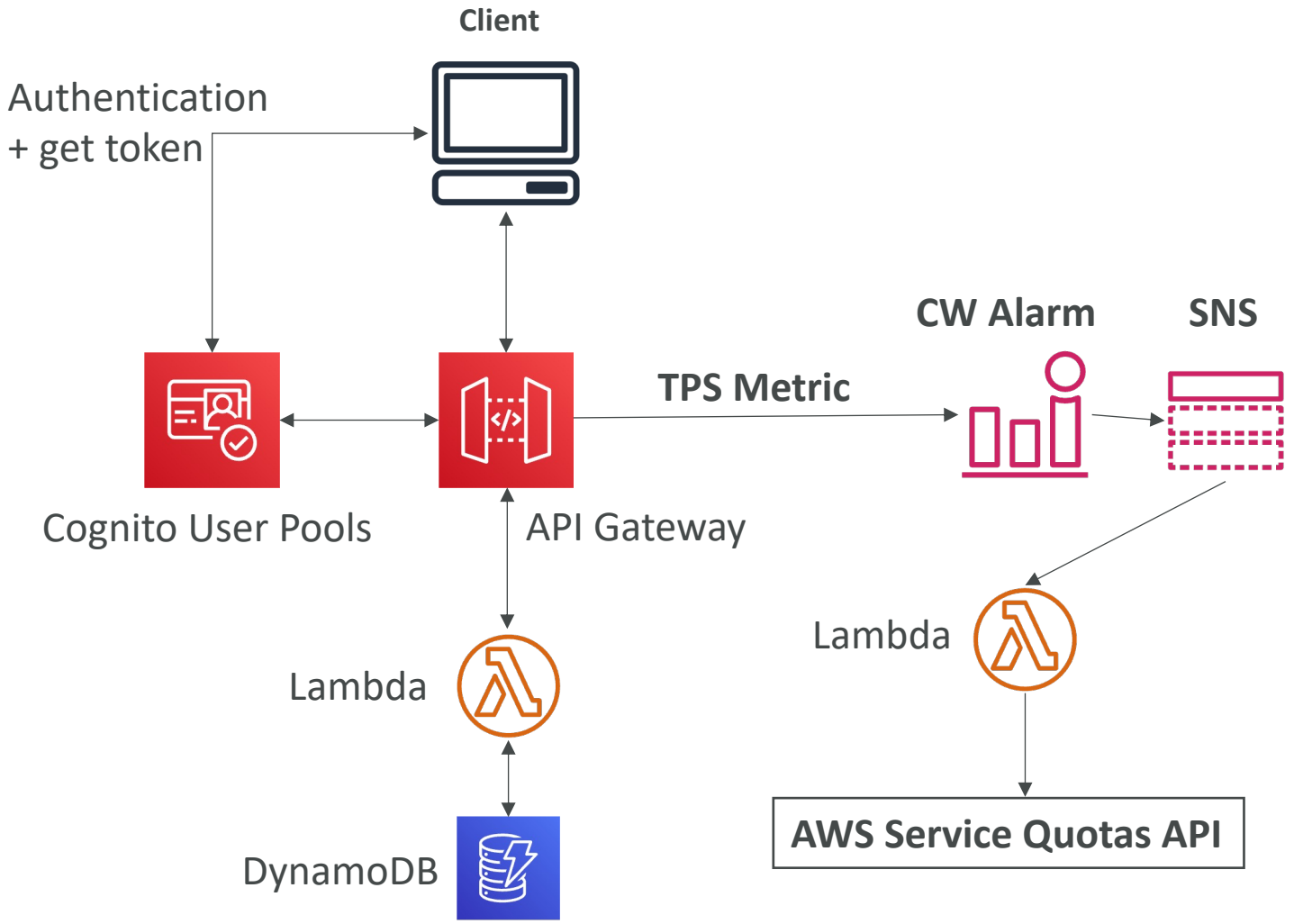
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Option B

- Configure notification alerts for the limit of transactions per second on the API Gateway endpoint and create a Lambda function that will increase this limit, as needed.

Option B would work if we were receiving 429 errors
429: Quota exceeded, Throttle



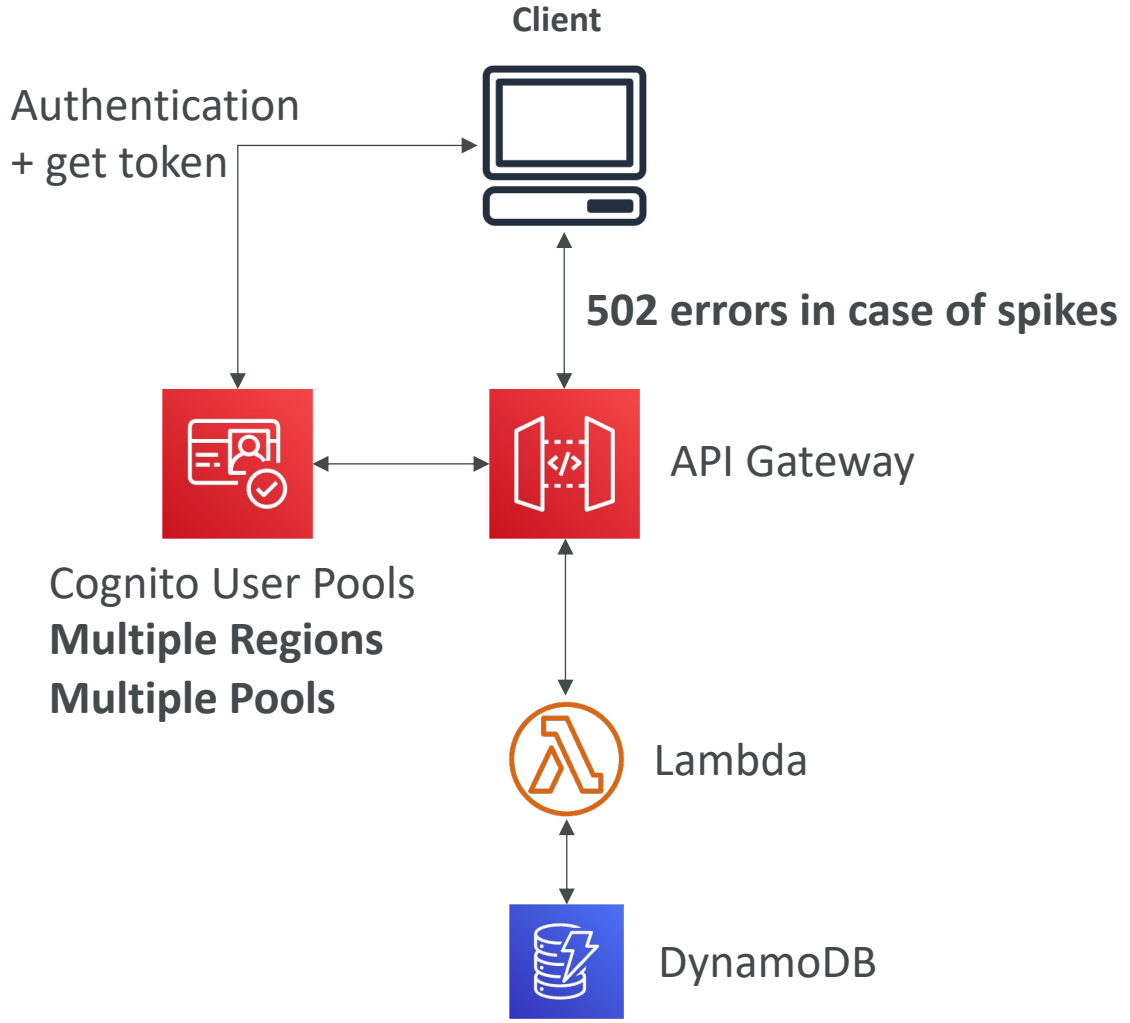
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Option C

- Shard users to Amazon Cognito user pools in multiple regions to reduce user authentication latency.

Option C would be valid if we had performance issues At Cognito User Pools, but that's not the question

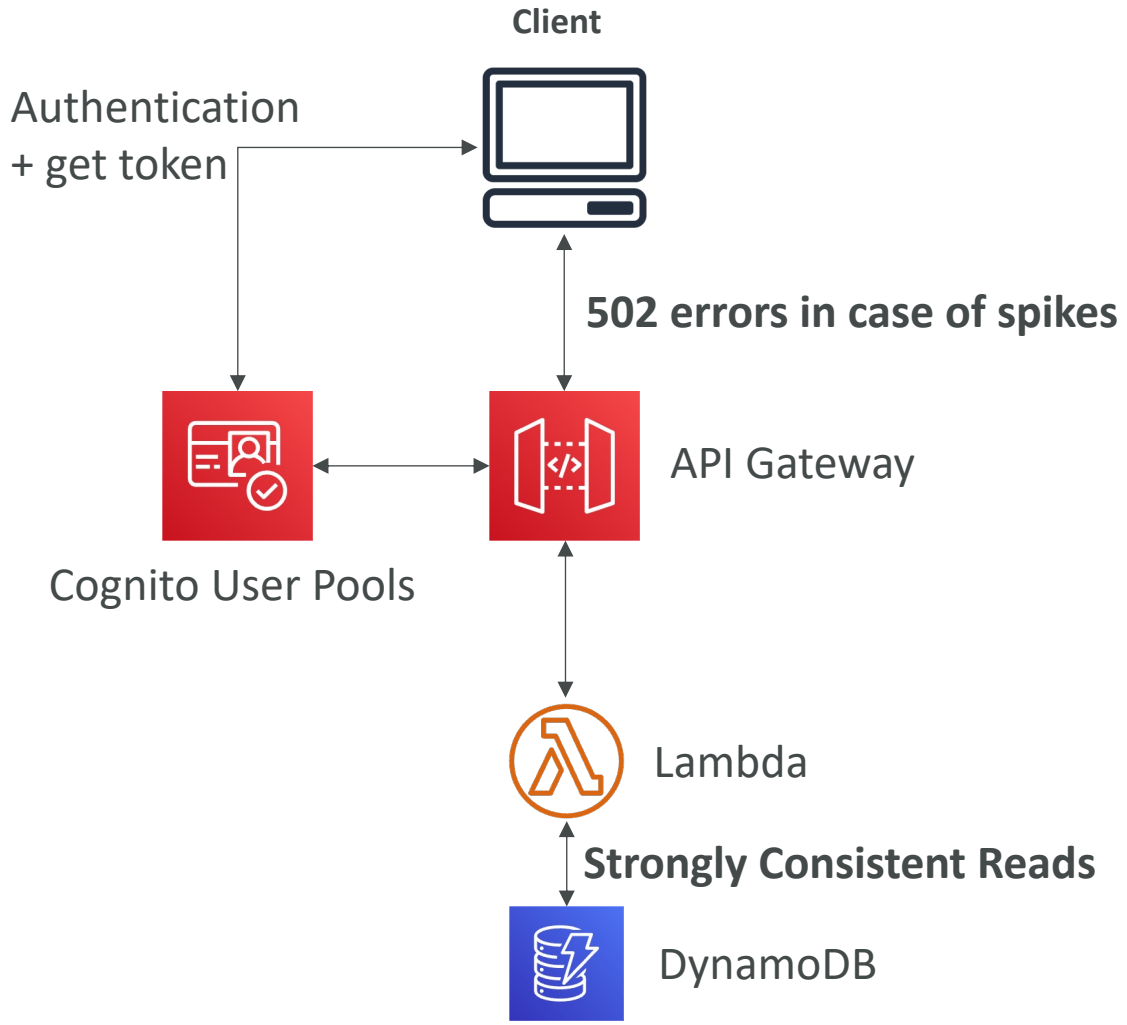


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Option D

- Use DynamoDB strongly consistent reads to ensure the latest data is always returned to the client application.



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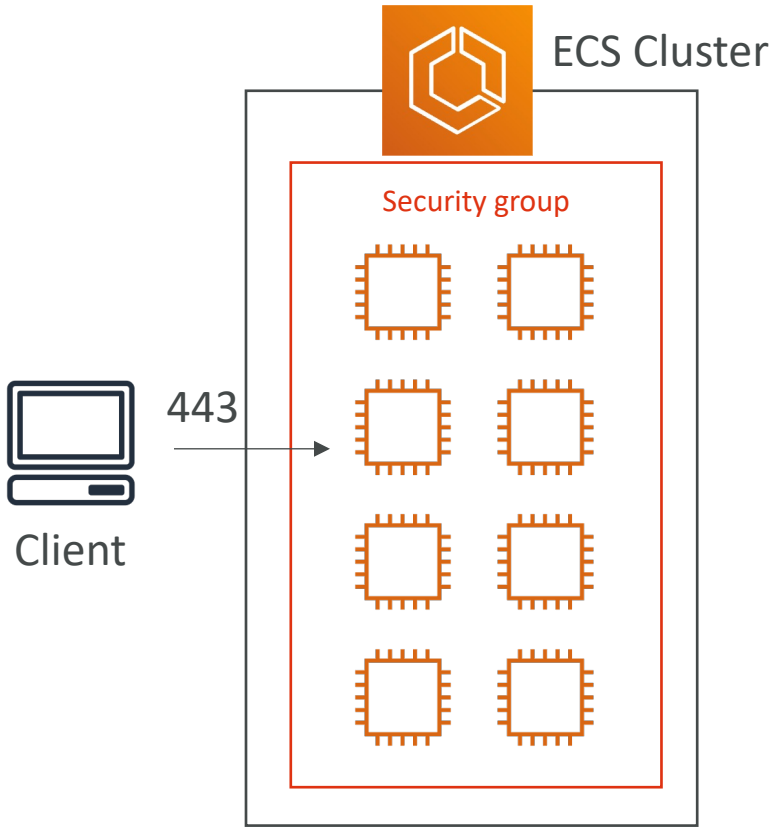
Question 5

- A company is launching a new web service on an Amazon ECS cluster. Company policy requires that the security group on the cluster instances block all inbound traffic but HTTPS (port 443). The cluster consists of Amazon 100 EC2 instances. Security engineers are responsible for managing and updating the cluster instances. The security engineering team is small, so any management efforts must be minimized.
- How can the service be designed to meet these operational requirements?

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Question 5 – Architecture

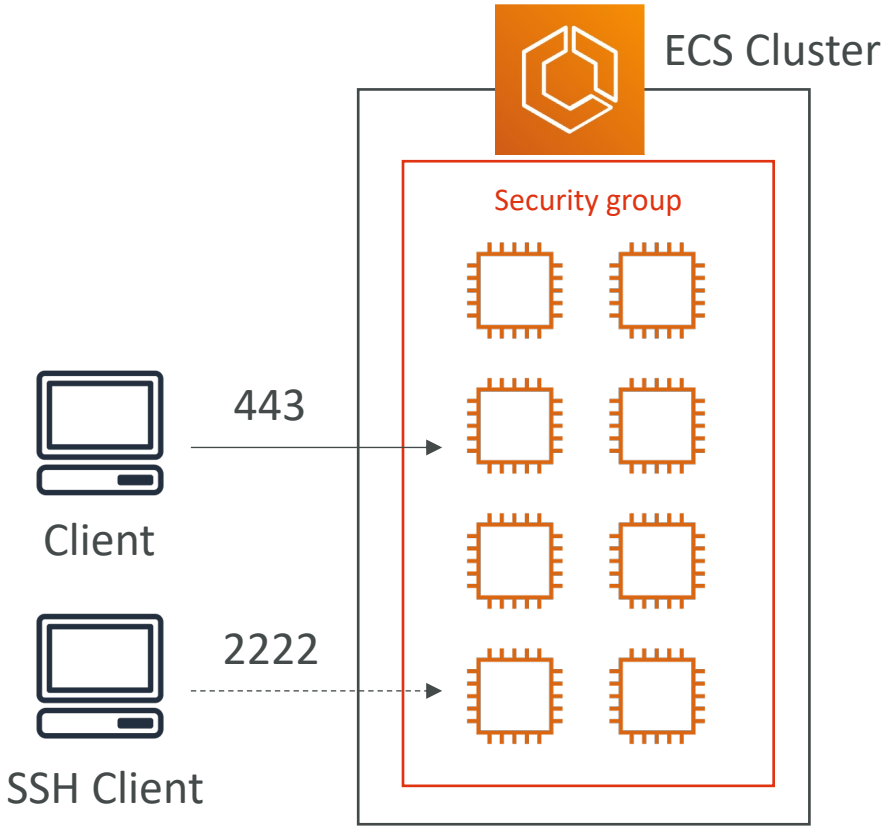


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Option A

- Change the SSH port to 2222 on the cluster instances with a user data script. Log in to each instance using SSH over port 2222

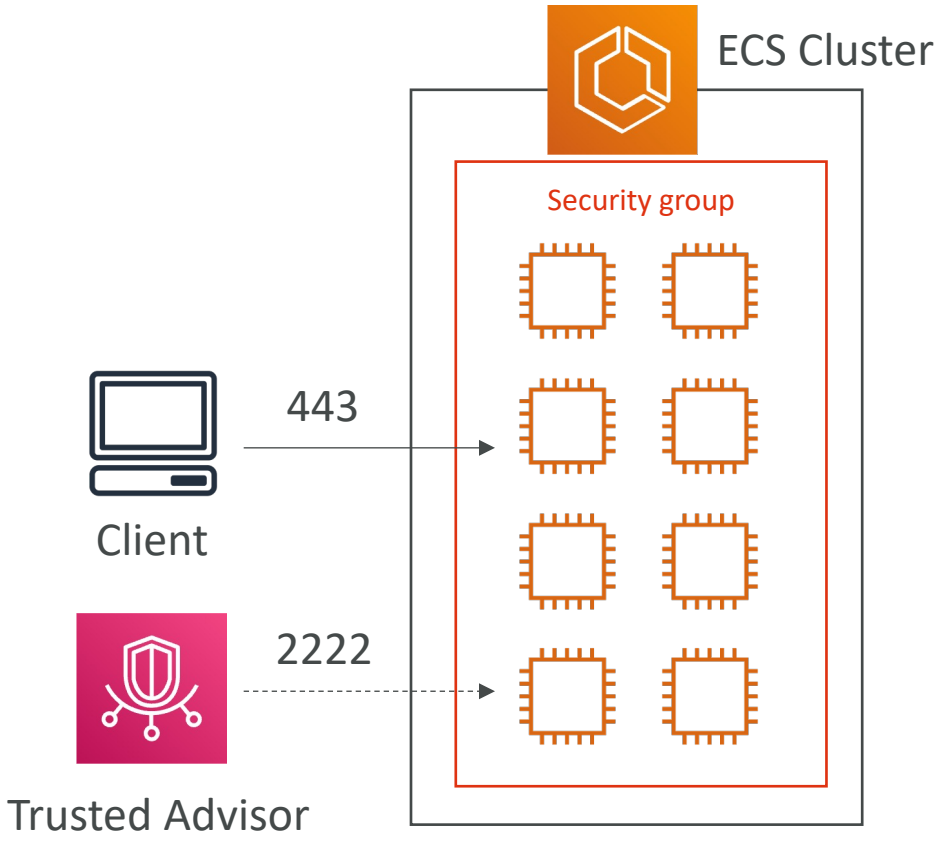


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Option B

- Change the SSH port to 2222 on the cluster instances with a user data script. Use AWS Trusted Advisor to remotely manage the cluster instances over port 2222



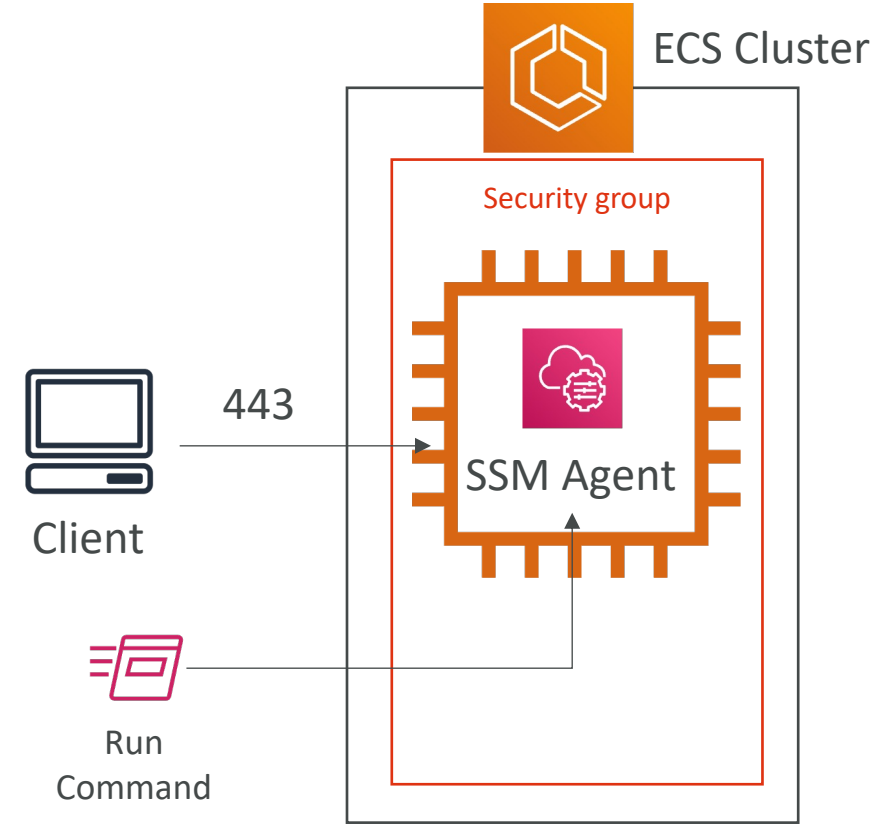
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Option C

CORRECT ANSWER

- Launch the cluster instances with no SSH key pairs. Use the Amazon EC2 Systems Manager Run Command to remotely manage the cluster instances

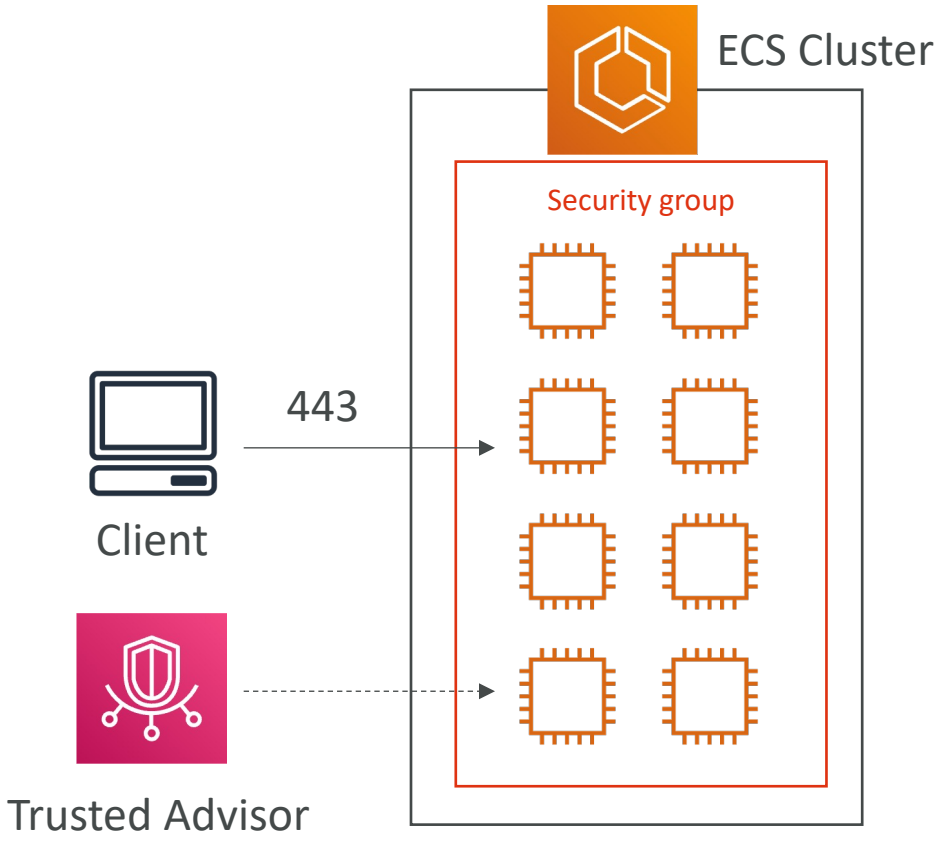


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Option D

- Launch the cluster instances with no SSH key pairs. Use AWS Trusted Advisor to remotely manage the cluster instances.



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Question 6

- A company has two AWS accounts: one for production workloads and one for development workloads. Creating and managing these workloads are a development team and an operations team. The company needs a security strategy that meets the following requirements:
  - Developers need to create and delete development application infrastructure.
  - Operators need to create and delete both development and production application infrastructure.
  - Developers should have no access to production infrastructure.
  - All users should have a single set of AWS credentials.
- What strategy meets these requirements?

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Option A

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each operator and developer and assign them to the development group.
- In the production account:
  - Create an operations IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each operator and assign them to the operations group.

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Option B

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group in the production account.
- In the production account:
  - Create an operations IAM group with the ability to create and delete application infrastructure.

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Option C

- In the development account:
  - Create a shared IAM role with the ability to create and delete application infrastructure in the production account.
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an operations IAM group with the ability to assume the shared role.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group.

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Option D

CORRECT ANSWER

- In the development account:
  - Create a development IAM group with the ability to create and delete application infrastructure.
  - Create an operations IAM group with the ability to assume the shared role in the production account.
  - Create an IAM user for each developer and assign them to the development group.
  - Create an IAM user for each operator and assign them to the development group and the operations group.
- In the production account:
  - Create a shared IAM role with the ability to create and delete application infrastructure.
  - Add the development account to the trust policy for the shared role.

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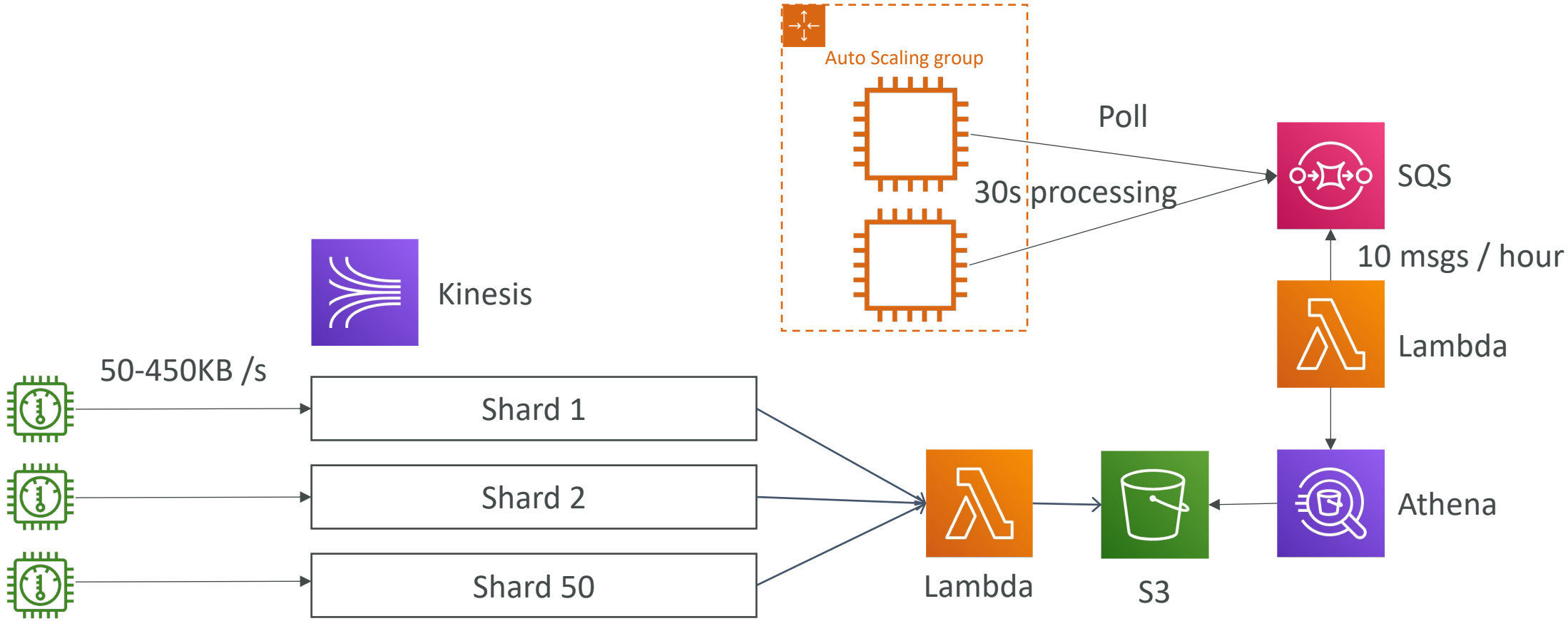
Question 7

- A solutions architect needs to reduce costs for a big data application. The application environment consists of hundreds of devices that send events to Amazon Kinesis Data Streams. The device ID is used as the partition key, so each device gets a separate shard. Each device sends between 50 KB and 450 KB of data per second. The shards are polled by an AWS Lambda function that processes the data and stores the result on Amazon S3.
- Every hour, an AWS Lambda function runs an Amazon Athena query against the result data that identifies any outliers and places them in an Amazon SQS queue. An Amazon EC2 Auto Scaling group of two EC2 instances monitors the queue and runs a short (approximately 30-second) process to address the outliers. The devices submit an average of 10 outlying values every hour.
- Which combination of changes to the application would MOST reduce costs? (Select TWO.)

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Question 7 – Architecture



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Options (choose 2)

A) Change the Auto Scaling group launch configuration to use smaller instance types in the same instance family.

B) Replace the Auto Scaling group with an AWS Lambda function triggered by messages arriving in the Amazon SQS queue.

C) Reconfigure the devices and data stream to set a ratio of 10 devices to 1 data stream shard.

D) Reconfigure the devices and data stream to set a ratio of 2 devices to 1 data stream shard.

E) Change the desired capacity of the Auto Scaling group to a single EC2 instance.

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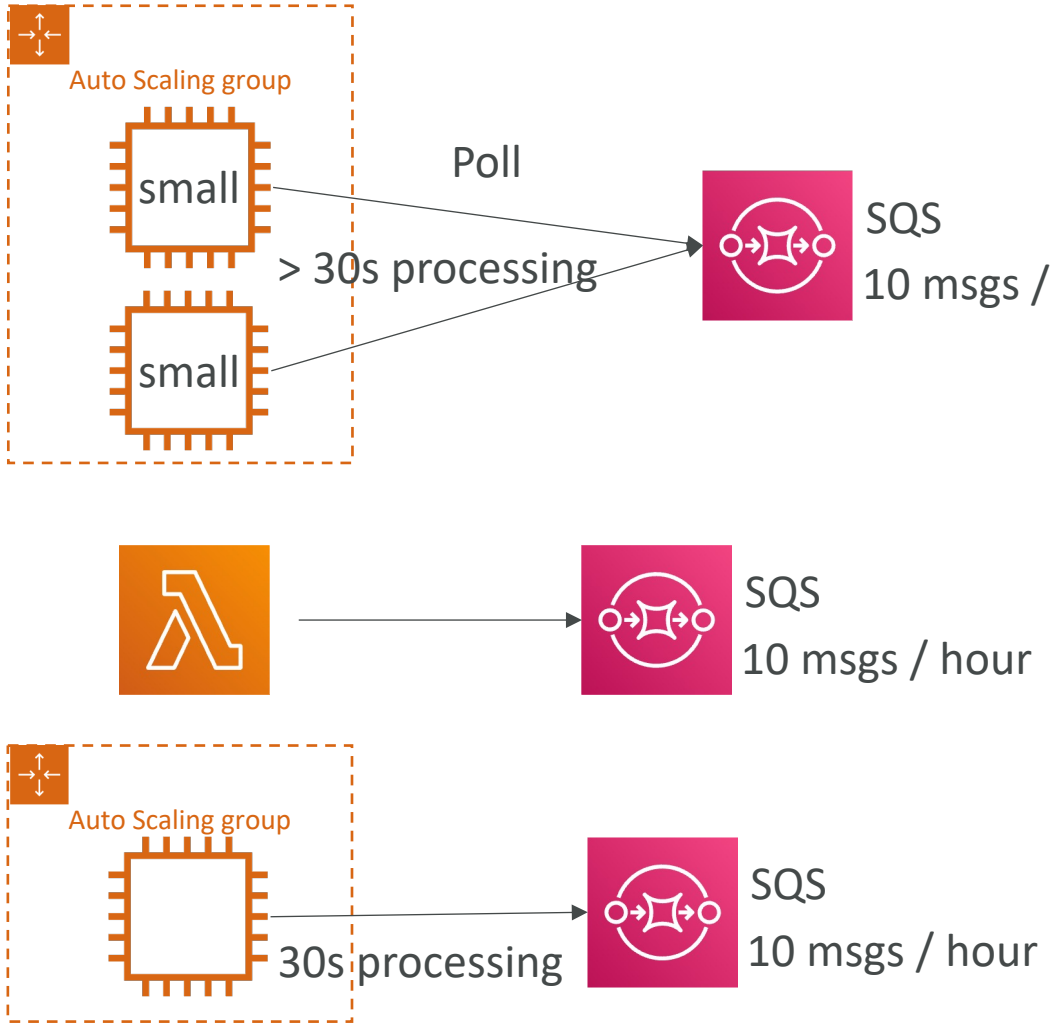
Option Group 1

CORRECT ANSWER B

A) Change the Auto Scaling group launch configuration to use smaller instance types in the same instance family.

B) Replace the Auto Scaling group with an AWS Lambda function triggered by messages arriving in the Amazon SQS queue.

E) Change the desired capacity of the Auto Scaling group to a single EC2 instance.



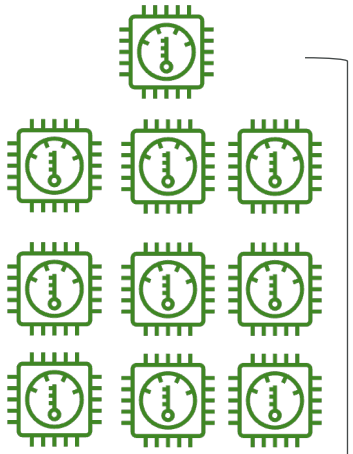
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Option Group 2

CORRECT ANSWER D

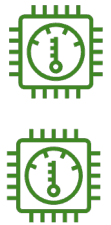
C) Reconfigure the devices and data stream to set a ratio of 10 devices to 1 data stream shard.



50-450KB /s x 10 = **500KB-4500KB / s**

Shard N

D) Reconfigure the devices and data stream to set a ratio of 2 devices to 1 data stream shard.



50-450KB /s x 2 = **100KB-900KB / s**

Shard N

Each shard has a limit of 1MB/s = 1000KB/s

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Question 8

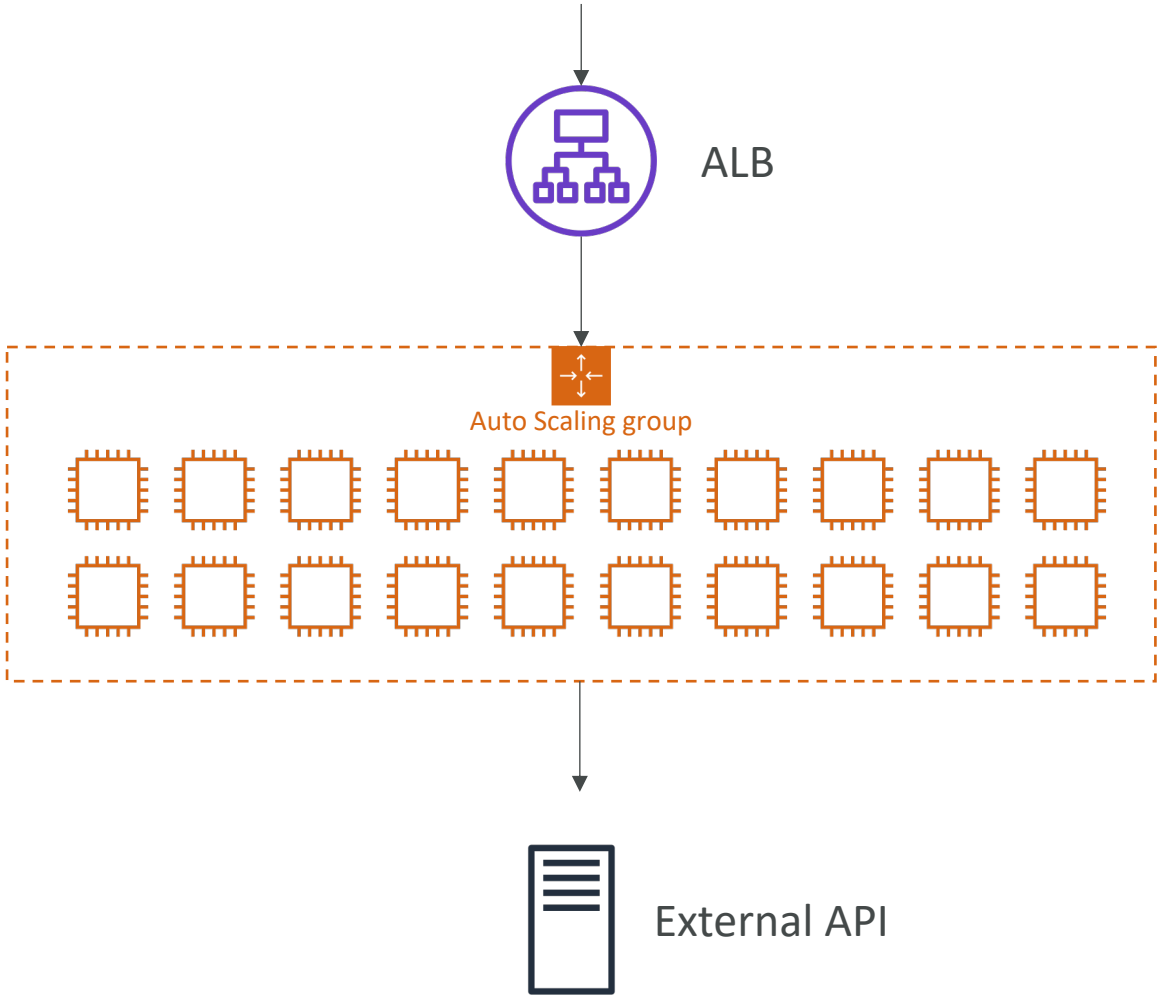
- A company operates an ecommerce application on Amazon EC2 instances behind an ELB Application Load Balancer. The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. After an order is successfully processed, the application immediately posts order data to an external third-party affiliate tracking system that pays sales commissions for order referrals. During a highly successful marketing promotion, the number of EC2 instances increased from 2 to 20. The application continued to work correctly, but the increased request rate overwhelmed the third-party affiliate and resulted in failed requests.

- Which combination of architectural changes could ensure that the entire process functions correctly under load? (Select TWO.)

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Question 8 – Architecture



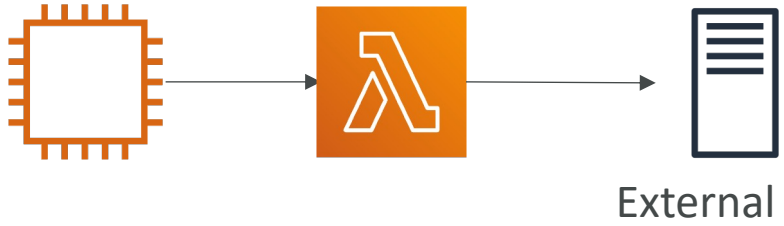
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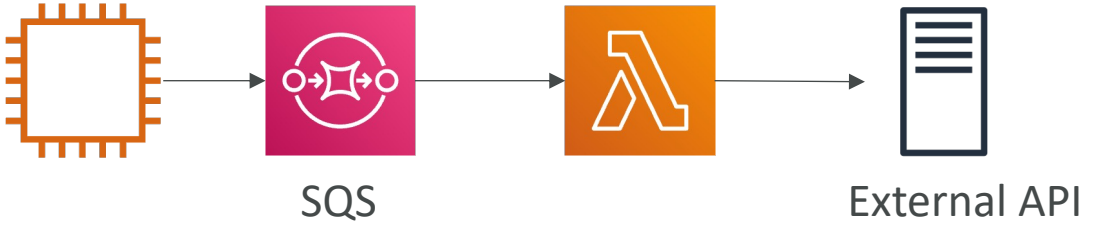
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Option Group 1

CORRECT ANSWER B

- A) Move the code that calls the affiliate to a new AWS Lambda function. Modify the application to invoke the Lambda function asynchronously.
- B) Move the code that calls the affiliate to a new AWS Lambda function. Modify the application to place the order data in an Amazon SQS queue. Trigger the Lambda function from the queue.





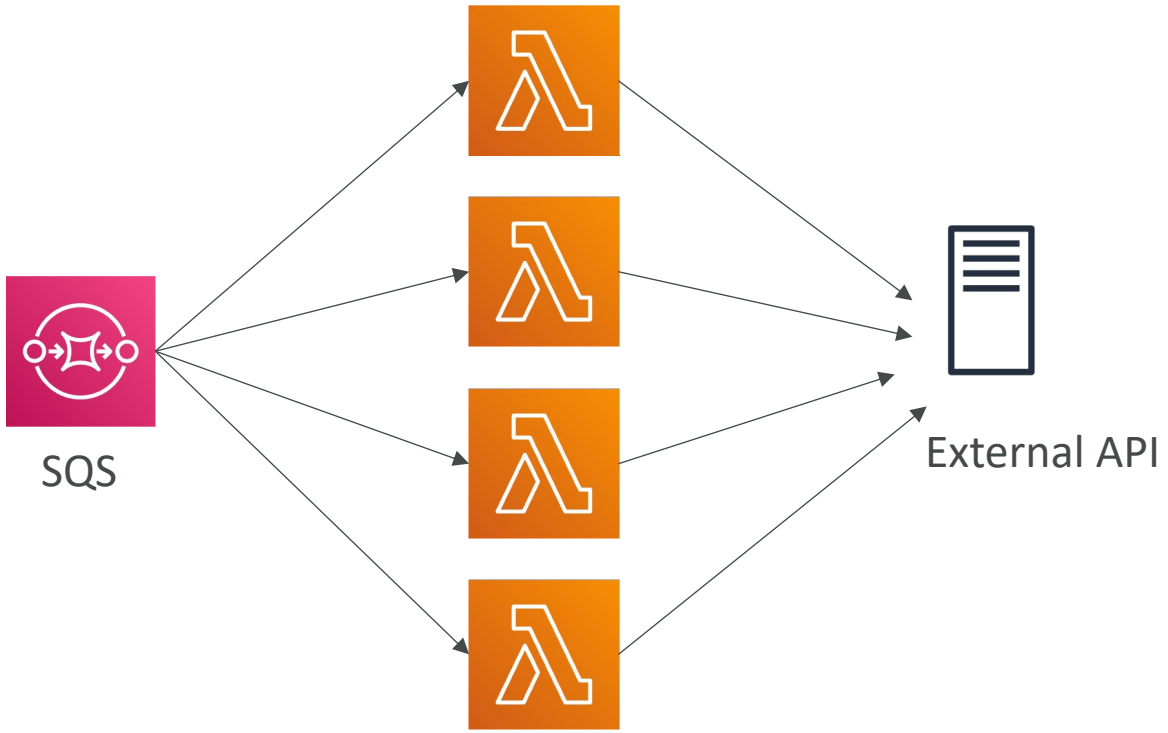
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Option Group 2

CORRECT ANSWER D

- C) Increase the timeout of the new AWS Lambda function.
- D) Adjust the concurrency limit of the new AWS Lambda function.
- E) Increase the memory of the new AWS Lambda function.



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Question 10

- A company has deployed a multi-tier web application in the AWS Cloud. The application consists of the following tiers:
  - A Windows-based web tier that is hosted on Amazon EC2 instances with Elastic IP addresses
  - A Linux-based application tier that is hosted on EC2 instances that run behind an Application Load Balancer (ALB) that uses path-based routing
  - A MySQL database that runs on a Linux EC2 instance

- All the EC2 instances are using Intel-based x86 CPUs. A solutions architect needs to modernize the infrastructure to achieve better performance. The solution must minimize the operational overhead of the application.

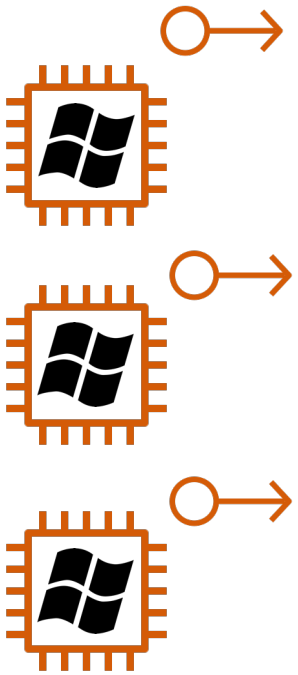
- Which combination of actions should the solutions architect take to meet these requirements? (Select TWO.)

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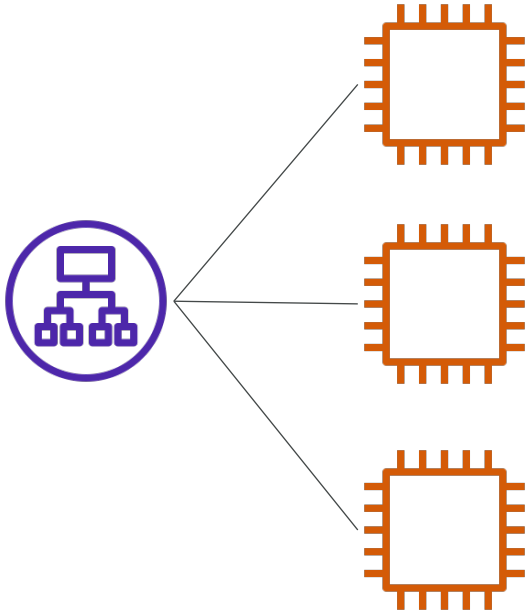
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Question 10 - Architecture

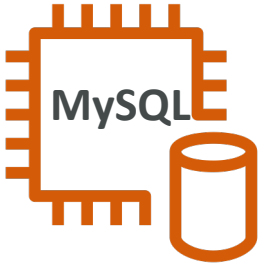
Web Tier



Application Tier



Database Tier



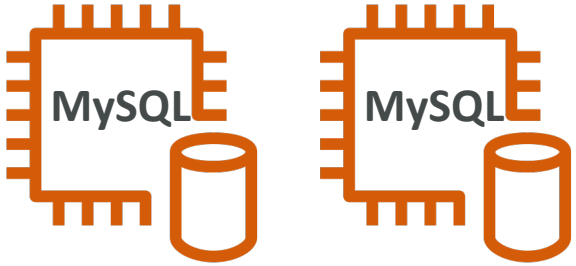
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Option Group 1

CORRECT ANSWER C

- A) Run the MySQL database on multiple EC2 instances.



- C) Migrate the MySQL database to Amazon Aurora Serverless.



- D) Migrate all EC2 instance types to Graviton2.
  - Not available for Windows Instances ✗

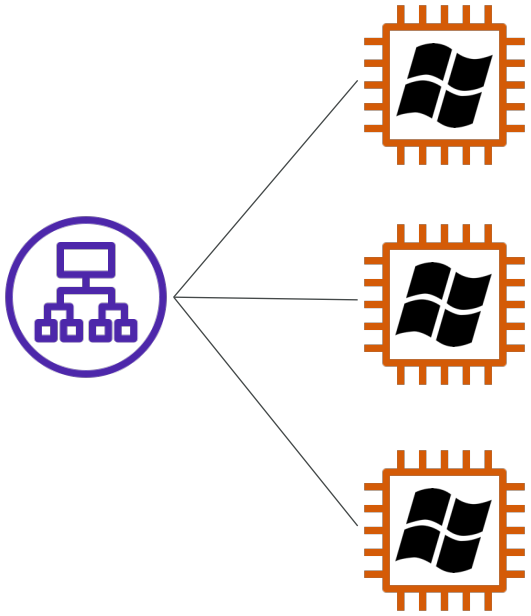
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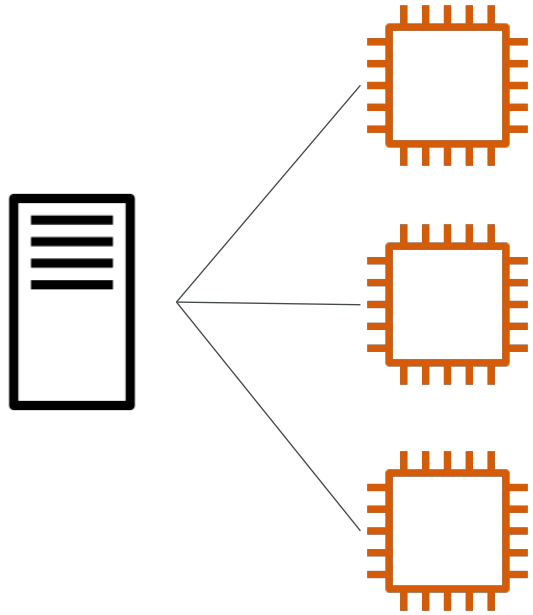
Option Group 2

CORRECT ANSWER B

- B) Place the web tier instances behind an ALB.



- E) Replace the ALB for the application tier instances with a company-managed load balancer.



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Your AWS Certification journey

Foundational

Knowledge-based certification for
foundational understanding of AWS Cloud.
No prior experience needed.



Associate

Role-based certifications that showcase your knowledge
and skills on AWS and build your credibility as an AWS Cloud professional.
Prior cloud and/or strong on-premises IT experience recommended.



Professional

Role-based certifications that validate advanced skills
and knowledge required to design secure, optimized,
and modernized applications and to automate processes on AWS.
2 years of prior AWS Cloud experience recommended.



Specialty

Dive deeper and position yourself as a trusted advisor to your
stakeholders and/or customers in these strategic areas.
Refer to the exam guides on the exam pages for recommended experience.



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AWS Certification Paths – Architecture

Architecture

Solutions Architect

Design, develop, and manage cloud infrastructure and assets, work with DevOps to migrate applications to the cloud



optional for IT/ cloud professionals

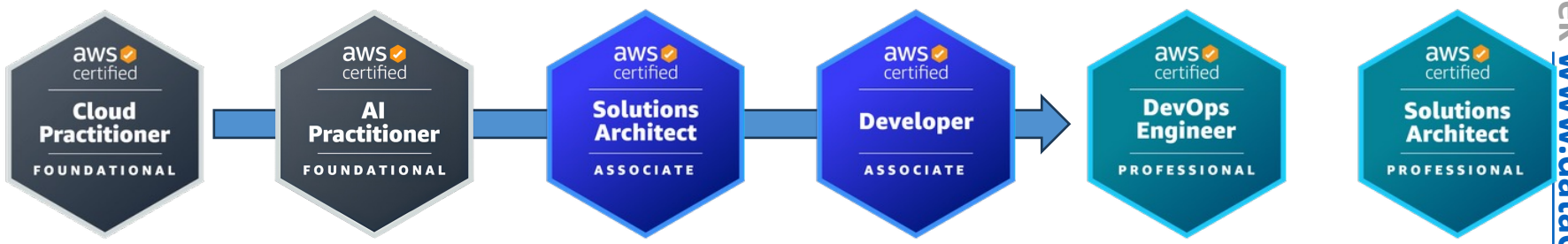
recommended for IT/cloud professionals to leverage AI

Dive Deep

Architecture

Application Architect

Design significant aspects of application architecture including user interface, middleware, and infrastructure, and ensure enterprise-wide scalable, reliable, and manageable systems



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Dive Deep

https://d1.awsstatic.com/training-and-certification/docs/AWS_certification_paths.pdf

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AWS Certification Paths – Operations

Operations

Systems Administrator

Install, upgrade, and maintain computer components and software, and integrate automation processes



optional for IT/ cloud professionals



Dive Deep

Operations

Cloud Engineer

Implement and operate an organization's networked computing infrastructure and Implement security systems to maintain data safety



optional for IT/ cloud professionals



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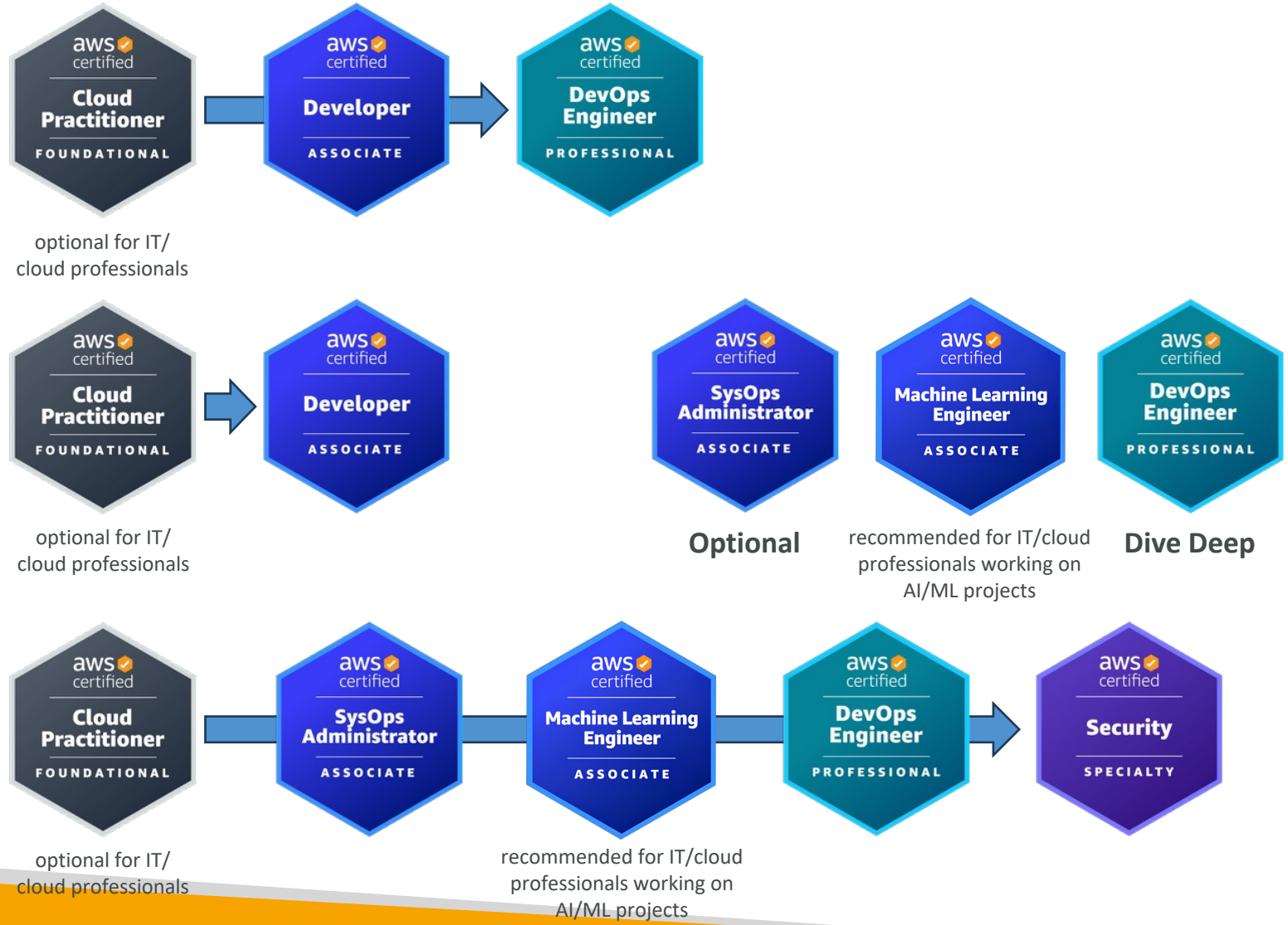
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AWS Certification Paths – DevOps

DevOps
Test Engineer
Embed testing and quality best practices for software development from design to release, throughout the product life cycle

DevOps
Cloud DevOps Engineer
Design, deployment, and operations of large-scale global hybrid cloud computing environment, advocating for end-to-end automated CI/CD DevOps pipelines

DevOps
DevSecOps Engineer
Accelerate enterprise cloud adoption while enabling rapid and stable delivery of capabilities using CI/CD principles, methodologies, and technologies



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AWS Certification Paths – Security

Security

Cloud Security Engineer

Design computer security architecture and develop detailed cyber security designs. Develop, execute, and track performance of security measures to protect information



optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

Security

Cloud Security Architect

Design and implement enterprise cloud solutions applying governance to identify, communicate, and minimize business and technical risks



optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

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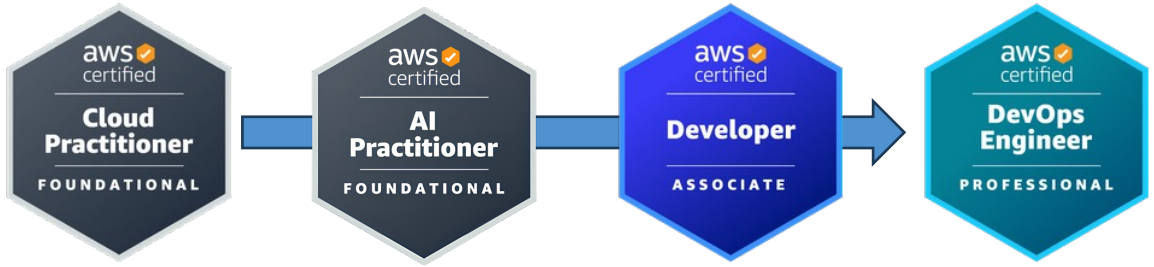
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AWS Certification Paths – Development & Networking

Development

Software Development Engineer

Develop, construct, and maintain software across platforms and devices



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Networking

Network Engineer

Design and implement computer and information networks, such as local area networks (LAN), wide area networks (WAN), intranets, extranets, etc.



optional for IT/ cloud professionals

Dive Deep

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AWS Certification Paths – Data Analytics & AI/ML

Data Analytics

Cloud Data Engineer

Automate collection and processing of structured/semi-structured data and monitor data pipeline performance



optional for IT/ cloud professionals

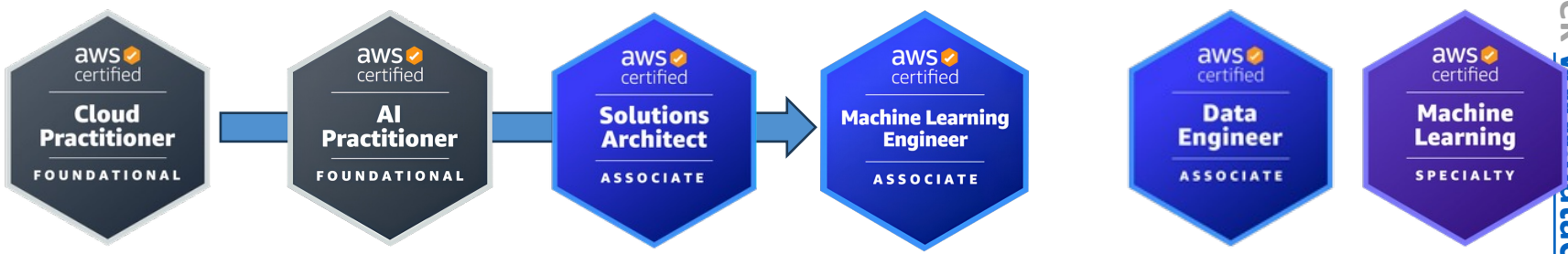
recommended for IT/cloud professionals working on AI/ML projects

Dive Deep

AI/ML

Machine Learning Engineer

Research, build, and design artificial intelligence (AI) systems to automate predictive models, and design machine learning systems, models, and schemes



optional for IT/ cloud professionals

optional for AI/ML professionals

Dive Deep

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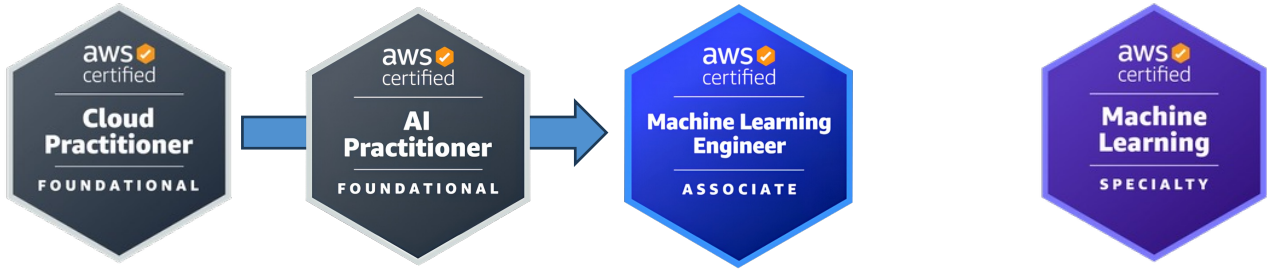
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AWS Certification Paths – AI/ML

AI/ML

Prompt Engineer

Design, test, and refine text prompts to optimize the performance of AI language models



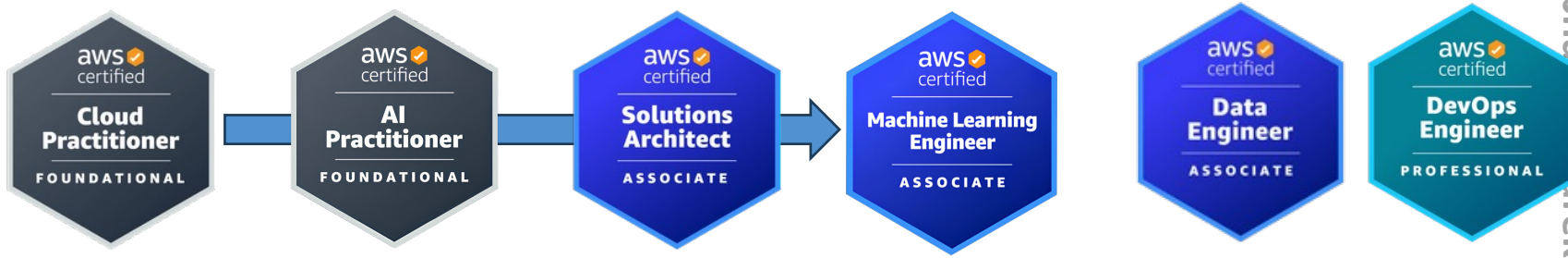
optional for IT/ cloud professionals

Dive Deep

AI/ML

Machine Learning Ops Engineer

Build and maintain AI and ML platforms and infrastructure. Design, implement, and operationally support AI/ML model activity and deployment infrastructure



optional for IT/ cloud professionals

optional for AI/ML professionals

AI/ML

Data Scientist

Develop and maintain AI/ML models to solve business problems. Train and fine tune models and evaluate their performance



optional for IT/ cloud professionals

optional for AI/ML professionals

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